

Quantifying Whether Virtual Reality Alters Juiciness Effects



UNIVERSITY OF
LINCOLN

Joel Carter

16598945

16598945@students.lincoln.ac.uk

School of Engineering and Physical Sciences
College of Health and Science
University of Lincoln

Submitted in partial fulfilment of the requirements for
the Degree of BSc (Hons) Games Computing

Supervisor: Dr. Craig Green

May 2025

Abstract

This paper aims to summarise the process of quantifying whether virtual reality alters juicy effects. An artifact containing three separate mini games will be developed to explore different effects on virtual reality from juiciness, defined as a reaction to player action to improve game feel, and will include concepts based on memory, motivation and competence. Each mini game will have a counterpart without juicy elements connected to items and player actions. All mini games will be designed with a score in mind to allow results to be assessed and compared, after which a Player Experience Index (PXI) will be completed to measure the overall quality of the artifact. This study aims to fulfill research into juiciness and virtual reality, a topic that has largely gone overlooked despite frequent calls for further study within related works, the intention being to draw parallels between the strengths of both juiciness and virtual reality in order to measure whether these concepts alter each others benefits in any meaningful ways. The outcome of this research may improve future virtual reality game design for recreational games and other forms including instructional and educational applications.

Table of Contents

1	Chapter 1: Introduction	1
1.1	Motivation	1
1.2	Project Structure	1
2	Chapter 2: Literature Review	2
2.1	Juiciness Definitions	2
2.2	Juiciness Interrelation	2
2.3	Virtual Reality Design	3
2.4	Virtual Reality and Immersion	3
2.5	Competence and Motivation	4
2.6	Aims and Objectives	4
2.7	Risk Analysis	6
3	Chapter 3: Requirements Analysis	7
3.1	Project Planning	7
3.2	Toolset	8
4	Chapter 4: Design & Methodology	8
4.1	Artefact Design	8
4.2	Testing	10
4.3	Research Involving Human Participation	12
5	Chapter 5: Implementation	15
5.1	XR Interactables	15
5.2	User Interface	15
5.3	Memory Game	17
5.4	Juiciness Implementation	20
5.5	Boring Game	22
5.6	Time Trail Game	24
6	Chapter 6: Results & Discussion	25
6.1	PXI Results	25
6.2	Quantitative Results	48
6.3	Discussion	51
7	Chapter 7: Conclusion	53
	References	54

1 Chapter 1: Introduction

1.1 Motivation

The concept of juicy feedback in game design, that being the responsive reaction to player actions, such as visual effects, audio cues and haptic responses, is widely recognised for enhancing player satisfaction and engagement in video games. This feedback creates a sense of responsiveness, making games feel more immersive and rewarding. However, as virtual reality grows as an industry and technology develops, the application of juicy feedback in VR remains largely unexplored, despite it's unique ability to provide immersive experiences.

Virtual reality as a medium which has the potential to amplify the effects of juicy feedback beyond what is achievable in traditional video games. For instance, Kao et al. (2024) demonstrated that in a VR game, juicy feedback significantly enhanced players' sense of competence, autonomy, relatedness, presence and enjoyment.

Despite these promising findings, there is a notable gap in literature regarding how VR specifically alters juicy feedback, particularly in relation to cognitive and emotional outcomes such as memory, motivation and competence. These aspects are important for enhancing player experience in games and also for improving engagement and learning outcomes in educational and instructional VR applications. The lack of comprehensive research in this area, coupled with frequent calls for further investigation within surrounding studies, highlights the need to quantify the relationship between virtual reality and juicy feedback.

Therefore this research seeks to provide a deeper understanding of how VR alters juicy effects. The findings are expected to contribute to both academic knowledge and practical game design, offering insights that can guide the creation of more enjoyable VR experiences.

1.2 Project Structure

The structure of the project will go as follows; A literature review will focus on the benefits of both VR and juicy effects to find the overlapping overall benefits of both concepts, Aims and objectives will clearly define the projects goals where both a VR application will be developed and it's juicy effects will be tested, A risk analysis will outline possible pitfalls and their mitigating factors that are specific to this particular project, The project plan methodology will provide that planning for the development of the project while outlining the tools identified to best develop the artifact and results gathering, an artefact will be developed with Lincoln Ethics Application System (LEAS) approved human participation in mind for results gathering through mini-game scores and a Player Experience Inventory (PXI). Results and discussion will show the findings of the projects aims and the success to which this project fulfilled it's intended goals and finally the conclusion to which an evaluation of the project findings, the concepts learned within the project and any plans for investigation in the future.

2 Chapter 2: Literature Review

This review reflects on the development of ideas surrounding the relationship between juiciness, player experience and virtual reality design, while developing juicy mini-games based on gathered academic knowledge for testing purposes. The aim being to better understand juiciness within the context of virtual reality, connecting existing understanding of this juicy game design relationship to allow better game design practices in the future within any virtual reality context.

2.1 Juiciness Definitions

According to Hicks et al. (2019), both academia and industry view juiciness as a means of improving the player experience, stating juice improves the three basic psychological needs for competence, autonomy and relatedness. However, Kao (2020) suggests that few empirical investigations have looked at how juiciness concretely impacts players, this is in part due to a lack of clarity on the definition for juice, Fabre et al. (2024) mentions two possible definitions, the first being amplification of feedback to understand mechanics, Pichlmair & Johansen (2021) states juicing amplification provides coherence between different feedback sources, and the second as an abundance of audiovisual feedback Hicks et al. (2019), as a result, with the virtual reality market likely to grow exponentially Shukla (2020), it is important to understand the design of virtual reality games, meaning further study of visual game elements is a worthwhile research endeavor Pichlmair & Johansen (2021), this may in turn provide additional context to defining juiciness.

There is some disagreement on a specific definition for juiciness Singhal & Schneider (2021), this study aims to keep a generalist definition, as such common factors between definitions must be found, Ballou et al. (2021) finds all definitions include an exaggeration of a reaction from an action, Kao et al. (2024) evaluates the discussion surrounding the juiciness definition, stating juiciness is commonly framed as a part of "game feel", in reference to Swink (2009). Understanding the relationship between exaggeration and game feel is the basis for Hicks et al. (2018), where 13 heuristics are used for "Juicy design", therefore this juicy definition, where juiciness is a reaction to player action to improve game feel, will be useful within the development of this study.

2.2 Juiciness Interrelation

Enjoyment in games is strongly tied to visual embellishments to a certain degree, Durmanova (2022), and juicy sound effects create a greater sense of presence Smets & Van der Spek (2021), these positive effects on the user experience appeal to the user's senses, an important design element to provoke an emotional response in virtual reality design discussed by Dozio et al. (2022), however, Many aspects of player experience are interrelated Ribelles et al. (2024), and may share common attributes, for example, Tao et al. (2021) mentions game designs focused on aesthetics of sense-pleasure, discovery, and narrative, juiciness as an example

of sense-pleasure, rewards the player for engaging with actions in the world Smets & Van der Spek (2021), this shows juiciness acting within a larger system to connect mechanics and game aesthetics.

Therefore, it is important to understand how enjoyment and presence connect to each other within game design, Yang & Zhang (2022) concludes combining presence and flow is a way of facilitating enjoyment where flow is dependent on a user’s own will, emotion and disposition, factors affected by presence. Bodzin et al. (2021) finds flow to be correlated with a success of a VR activity, Frommel et al. (2021) backs their research on the grounds that ‘success in non-game contexts is usually associated with positive experiences’, within a gaming context, this ‘can be associated with game enjoyment’, this connects to juiciness where Hicks et al. (2019) finds juiciness influences perceived competence and engagement, a users own belief of success therefore facilitates flow, improving presence and enjoyment.

2.3 Virtual Reality Design

Developing within virtual reality has unique challenges that allow for unique game designs, Maroungkas et al. (2023) asserts it is progressively crucial for educators to learn to apply theories related to designing VR, Dozio et al. (2022) presents methodology for virtual reality designing, this selects a real emotion, finds which design elements contribute towards that emotion, then connects these elements together, in this way the creation of a virtual space is predicated on the intended emotion, this connects virtual reality design to player experience, Abeele et al. (2020) states Games User Researchers aim to understand how specific game choices lead to specific emotional responses, Makantasis et al. (2021) refers to this response as arousal, finding it can be predicted with high accuracy as long as an audiovisual feed captures the game play context, this context is highly dependent on the game’s design however. Somarathna et al. (2022) reviews papers incorporating VR in emotional studies, an upward trend was found towards VR usage from 2011 to 2021, juicy design should therefore highly focus on emotional response within VR.

2.4 Virtual Reality and Immersion

Virtual reality naturally brings a sense of immersion to a user, Frost et al. (2022) restoration and stress recovery was improved from exposure to videos of nature, Browning et al. (2020) found this effect within VR too, where virtual reality had lessened yet similar effects to being outside when simulating nature, this shows virtual realities natural ability to immerse users, Petersen et al. (2022) also finds this, where immersion and interactivity are shown to significantly affect learning, both of which were bolstered within virtual reality for educational purposes, this natural immersion advantage may be bolstered by the effects on users from juiciness, Luz et al. (2024) states juicy design enhances a game-play experience by inciting feelings of immersion, Durmanova (2022) explains juicy game elements create a larger aesthetic appeal, accentuating player actions,

adding more meaning and immersion in games. Perhaps it is possible juiciness has additional effects within virtual reality, it is clear both concepts contain properties that both improve immersion and perhaps when combined, these can become greater than the sum of each.

2.5 Competence and Motivation

Competence can be an important aspect for user engagement, Li et al. (2024) finds trivial or simple tasks are not intrinsically motivating, competence appears to motivate and this feeling cannot be achieved without challenge, this is backed up by Vo et al. (2022) found individuals with high competence tended to be less motivated at work, however more motivators beyond challenge may exist, Huang et al. (2021) received feedback from a participant connecting immersion motivation and engagement through novelty, virtual reality is more likely to be more novel to users when compared to other games media, this could have a surprising effect on users regarding motivation, naturally novel tasks can be intrinsically more difficult due to a lack of time to grow skill, González-Cutre et al. (2025) found novelty frustration positively related to negative consequences with physical education, lack of experience can lead to a lower challenge needed to gain a flow state within games, leading to frustration, motivation and competence, this can mean challenge effecting motivation is a two way relationship where motivation can also effect competence, requiring more challenge, as discussed by Sheeran et al. (2021) where competence and motivation play an important role in self determination theory.

2.6 Aims and Objectives

Aims Aim 1: Develop a VR application which holds multiple mini-games that are each designed to accentuate a chosen effect of juiciness.

Aim 2: Measure if juiciness was altered in a user study, when played within VR versus a desktop version, recording whether VR had an influence if any.

Objectives Objectives for Aim 1:

- Review which benefits of juiciness will be the focus for each mini-game.
- Design the way each mini-game will measure its given effect.
- Find open source tools to improve development, particularly for creating juicy effects.
- Develop at least four mini-games, each as a self contained experience.
- Develop a VR and desktop version of the application.
- Store player results of each mini-game for future comparison.
- Iteratively test and develop to be ready for study within four months.

Objectives for Aim 2:

- Ensure proper research ethics practices are followed.
- Research scientific heuristics for measuring effects for each mini-games focus.
- Develop a questionnaire for participants to compare VR and desktop overall experiences.
- Gather specific data from within each mini-game related to a juicy effect.
- Review each juicy effect to determine how virtual reality influenced them.
- Provide possible explanations for the results of findings and any surprises.
- Process the findings of participant study within 6 months.

2.7 Risk Analysis

Risk Analysis Risk 1: Poor Game Design

Each mini-game will be designed with its associated juiciness benefit in mind, a poorly designed mini-game may lead to meaningless results.

Mitigation:

With proper design backed by scientific data, juiciness properties can be isolated as with past studies. These juicy effects have been isolated already, allowing for focus on design methods, for example ?, discussing player engagement, concludes prioritising the three psychological needs during the design process should ensure player engagement is one mini-games design process focus.

Risk 2: VR and Juicy Effect Obfuscation

When comparing juicy effects within the context of virtual reality, it's possible any changes could be obscured by the change in medium, computer versus VR headset.

Mitigation:

Context must be given to show effects of juiciness, past data like ? related to just VR and juiciness without VR ? will allow comparison to show whether these two elements together caused interesting juicy and VR interaction, rather than simply combining each effect.

Risk 3: VR Safety Precautions

When using VR, safety risks, where a player exists in one space but sees another, can cause damage to equipment or injury.

Mitigation:

Ensuring players are seated while playing and Not including mechanics which require fast movement such as throwing or punching will reduce risk. Each mini-game should also account for different movement types that don't include physical movement like joystick movement. Teleportation movement options should reduce motion sickness, as well as improving presence, as suggested by ?.

Risk 4: Managing the Effects of Player Skill

Participant experience, gained from playing mini-games could translate into increased competency and improved results, should the participant play both the computer and VR versions.

Mitigation:

Different sets of participants should be used to compare results for the computer and VR versions of each mini-game, allowing each participant to access each experience with fresh eyes and without familiarity, this removes risk to result data with the intention to remove external factors from effecting game result data and participant questionnaire answers.

3 Chapter 3: Requirements Analysis

3.1 Project Planning

The development process needed to be efficient while maintaining all goals necessary for proper testing and result efficacy. Since development time frames were known and self motivated without external clients, consideration into all aspects of the game requirement were set prior to development, meaning an agile development approach would not be necessary, A Gannt chart was instead created following a waterfall model, this gave time to consider game design and to maximise the desired juiciness properties, allowing for smooth development of the artifact as well as understanding for the best testing methodology, generating a clear project direction. With an understanding of desired outcomes an ethics application was also completed and approved early on in the development timeline, this alleviated pressure on the design process since the project path had clearly been explained and approved and multiple drafts of ethics approval would have reduced certainty on the final project results and methodology requirements. The end result of planning

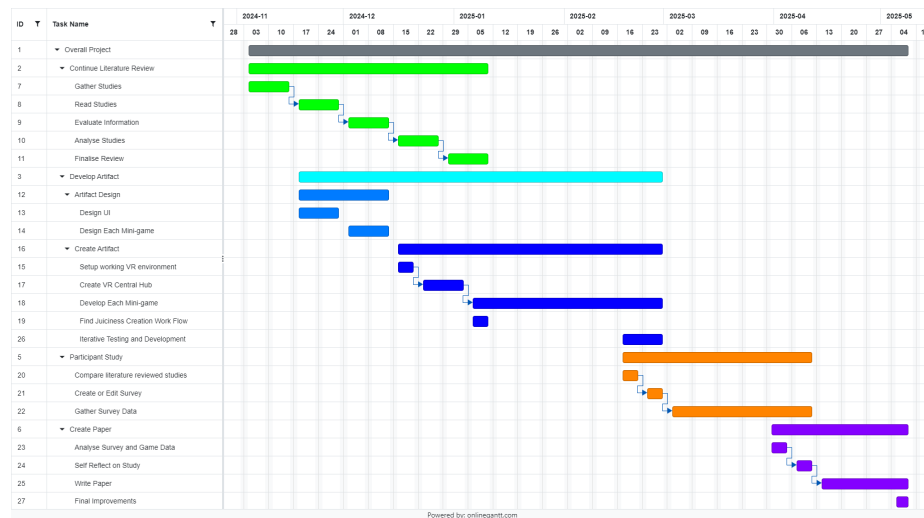


Fig. 1. This Gannt shows the development process of the project, starting 3rd of November 2024 and ending 8th May 2025, The VR game finished development aim is the 28th of February 2025 leaving ample time for data collection.

3.2 Toolset

To develop the artifact Unity was used, the unity XR Interaction Toolkit was especially useful for creating a basis for a working virtual world, this was an excellent starting point for further development as well as a great basis to learn virtual reality develop in general. The second reason was prior virtual reality experience, learning new development styles and mediums becomes easier within high familiarity environments, therefore Unity was the natural engine decision.

Another aspect to the use of Unity was the large number of free assets available online, although owned by epic games, fab.com Epic-Games (2025) was used for free assets including textures and the application of juicy events within the game, these free assets were easily transferrable to unity and quickened the development pipeline. Pre-made free juicy effects were especially useful as multiple juicy particles were provided and became a basis for the juice objects used to provide the separation between the juicy and non juicy mini-game variants.

Lastly Unity is not a light weight engine, however compared to unreal engine the minimum computer specification requirements are much friendlier, giving Unity the edge over the other potential solution. If this project artifact was much larger and requiring of additional feature providing within the base unreal projects then this would have been a suitable solution, however the performance issues caused by Unreal could have taken away from the player experience and may have affected final results.

4 Chapter 4: Design & Methodology

4.1 Artefact Design

When developing the artefact, a structured design approach is essential to ensure that the three mini-games effectively evaluate the impact on memory, motivation, and competence. The design process for this project involved laying out detailed plans for each mini-game, naming necessary object to understand a scope for required development time and planning, learning about virtual reality development, and developing the artifact itself.

The process began with creating plans for each mini-game within a game design document, each designed to focus on one key aspect of the player experience: memory, motivation or competence. These plans outlined the core mechanics, player interactions, and the specific juicy feedback elements that would enhance the experience in the juicy version of each game. All mini-games were designed with a scoring system in mind to allow for objective performance comparisons, aligning with the project's goal of quantifying juiciness.

Main Menu

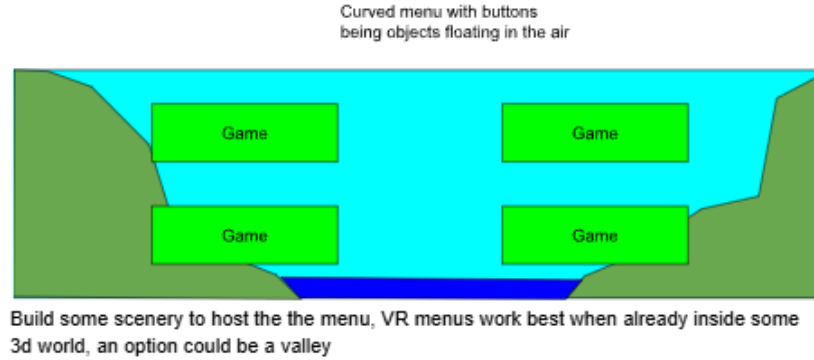


Fig. 2. An image creating during the planning phase for the main menu

Within each mini-game required objects were named to ensure organisation during development. This included interactive elements, user interface components, and assets. Naming objects early helped streamline asset creation and integration, reducing time spent on searching for or developing unnecessary resources. This approach allowed the consideration of game asset creation into the game development for planning and time management purposes ensuring a proper understanding of the artefact and projects current development position were understood and increased the clarity of development end goals.

Given the artefact was developed with VR in mind, learning about VR was a critical step, this involved studying Unity's VR-specific tools and techniques, implementing XR Interaction Toolkit, learning the pre-built components and uses, learning the creation of 3D UI within VR to create floating menus and screens as well as allowing player interaction with these UI elements using VR hand controller, learning the differences in which VR game design differs with regards to the position of the user's viewpoint, taking into account their perceived position within the world and the ways in which users interact with VR objects, all added to the unique challenges of learning virtual reality game design and development.

The design process ensured that the mini-games were well suited to investigate juiciness effects on player experience, while laying out clear definitions for required completion goals with regard to assets and the mini-games themselves.

4.2 Testing

Testing the artefact during development ensured it effectively evaluated the impact of juiciness on player experience, included within this were bugs that may detract from immersion related to presence and ensuring the games did not break or annoy participants to ensure data collection was smooth and easy process.

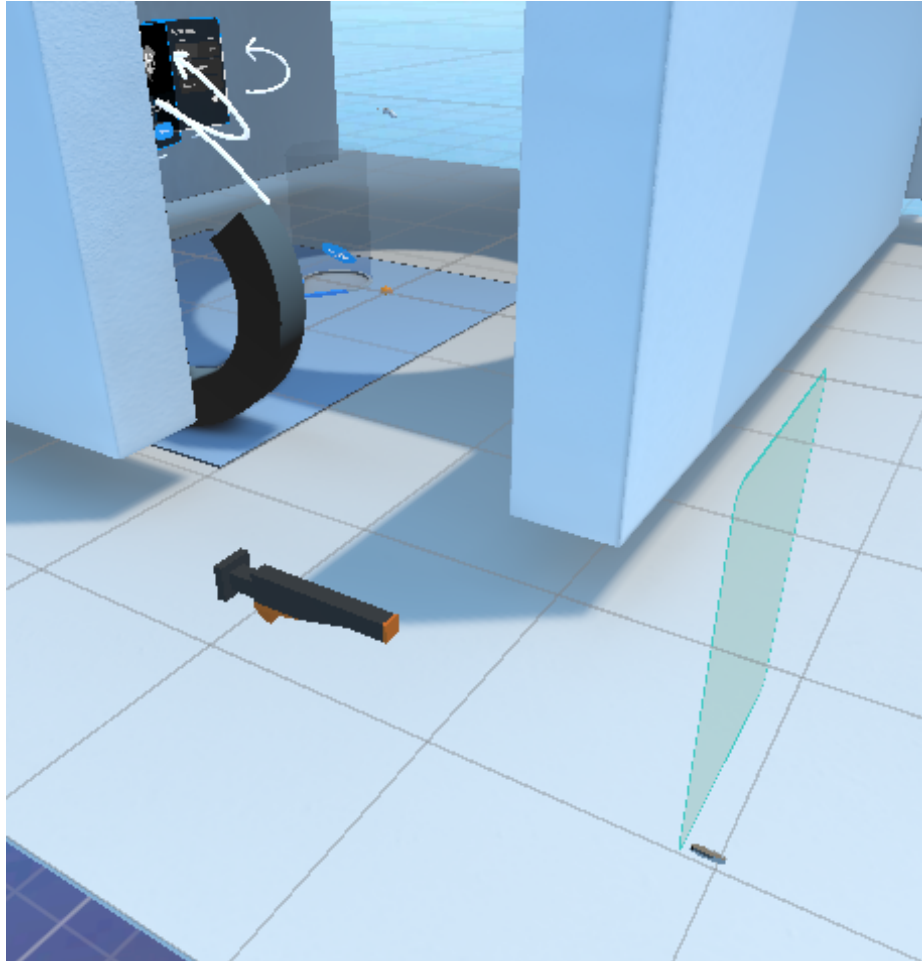


Fig. 3. This shows the XR Toolkit learning section and items added within that I used for training purposes

The largest change in testing for VR is the requirement for testing using a headset, this change can be mitigated by using an XR device simulator within the Unity XR toolkit, though this allowed for bug testing away from a head-

set and provided flexibility in development time, although the device simulator did not contribute to an understanding of the true game feel of each mini-game within VR. Because of this most testing was performed using the device simulator, however game design changes, and development milestones, including the completion of game mechanics, had to be tested within Lincoln University using the XR lab, since access to VR technology was not available elsewhere.

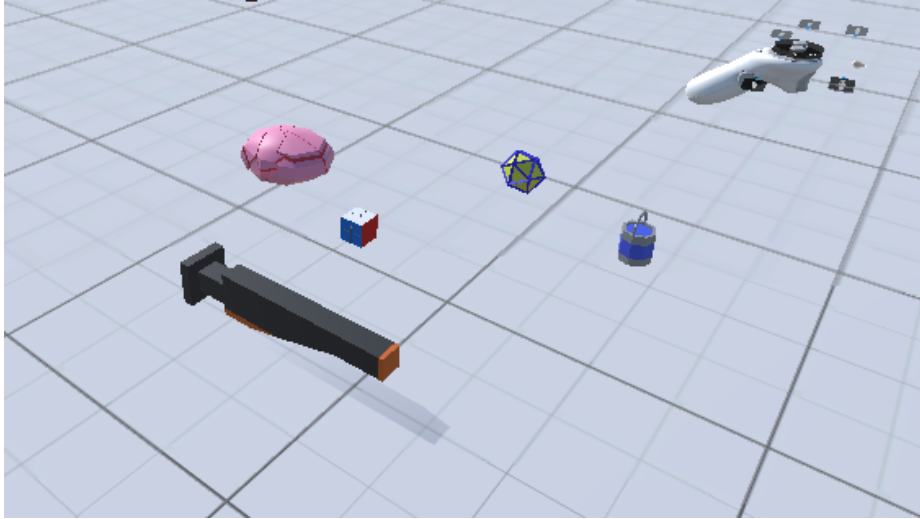


Fig. 4. This image shows objects used to play around with learning and developing interact-able objects

In addition, an understanding of the scope and ability of XR-Interactable objects within Unity's XR extension was critical for future planning and development, this took precedent over paper prototypes as non digital objects could not account for the pitfalls of VR development, as well as giving an understanding of what was possible to develop.

Creating objects to test the scope of their functionality provided an efficient way to learn VR development concepts and formulate the ideas which would grow to become the mini-games themselves. By creating and testing objects like interactive buttons, floating UI and guns, this method helped to integrate the learning process of this new development style as a whole, as well as build new assets that would ultimately become part of the mini-games in the future.

4.3 Research Involving Human Participation

Participant Recruitment Participants were required for this particular project as player experience needed to be assessed in order to gauge the success of this project and to evaluate the player experience gained when interacting with the artefact and support the research. Participants were recruited through family and university friends; but all participants used in the research were 18+ in accordance with research and legal guidelines, besides age, this project was accessible to all demographics and required no prior knowledge of previous research. To address concerns which arise from VR, participants were informed about the potential for motion sickness and were reassured they could withdraw if they became uncomfortable. Ethical approval was secured through Lincoln’s Ethical Approval System (LEAS), ensuring the study met UK standards for participant safety, informed consent, and data protection.

Evidence of Ethics Ethical procedures were prioritised throughout the study to safeguard participant confidentiality and ensure data security. All personal data gathered from participants, including their responses to the PXI questionnaire, were anonymised to prevent outside access and identification, thereby protecting their privacy in the study results. Participants were fully informed about the research purpose, a participation information sheet was provided with all the details of the study, as well as the risks involved, such as staying seated while playing, before agreeing to the study. Each participant provided written consent in paper form, which also informed of their right to withdraw from the study.

To maintain high ethical standards, the signed consent forms and PXI questionnaire answers, which included participants names and signatures, were stored securely. The PXI and consent forms were digitally scanned, then stored within a UoL OneDrive folder set for private, ensuring data was stored within the guidelines of LEAS and in accordance with data storage measure proposed during the LEAS ethics application.



Fig. 5. The files were stored on a University owned OneDrive account in private folders

Study Structure The study was conducted within the XR lab inside the Isaac Newton Building at the University of Lincoln, where participants attended individually scheduled sessions. Upon arrival, each participant was provided with a participant information sheet that detailed the study’s objectives and explained their role in the research. The information sheet also highlighted all the potential risks and assured participants they could take breaks or withdraw from the session without consequence. After reviewing the information sheet, participants who agreed signed a consent form.

Once consent was obtained, participants were fitted with a VR headset, care was put in to ensure a comfortable fit and time was allowed to minimise blurriness. The study compared player experiences between juicy and non juicy versions of three mini-games, each designed to assess a specific aspect of juiciness in memory, motivation and competence.

Participants were told which version of each mini-game they would play, this meant participants would play either juicy or non juicy a non juicy version of each game but not both, this minimised the effect on skill that previous experience in the other mini-game version could provide. After completing the three mini-games in an order of their choice, participants completed the Player Experience Inventory questionnaire to evaluate their experience.

In-between each mini-game participants were offered to have a short break and whether they needed water or fresh air. In some cases participants decided to end mini-games early without withdrawing which gave data related to cybersickness.



Fig. 6. A participant playing the memory game within the XR lab

Statistics Analysis Statistical analysis will be conducted using data collected from the 10 participants who played the three VR mini-games, there was an even number of participants that completed the juicy and the non juicy mini-game variants. Gathered information through the PXI shows the qualitative difference in experience for participants between variants while the mini-game result scores show the quantitative difference in performance. These results were then placed into graphs and compared to conduct final conclusions and answer the question posed in this paper.

5 Chapter 5: Implementation

5.1 XR Interactables

The initial task was to learn and implement the creation of interactive objects using Unity’s XR Interaction Toolkit to enable dynamic player interactions and juicy opportunities. This toolkit provides components such as XR Rig, XR Grab Interactable, and XR Direct Interactor, which facilitate natural interactions. For this project responsive feedback to player actions was particularly important so mastering this aspect of virtual reality development was key. This process also began within the memory mini-game, where learning the creation of different aspects of virtual reality objects also contributed to the number of existing objects within the memory game, simultaneously designing objects and developing the mini-game.

Grab-able objects were implemented allowing players to interact with them using VR controllers or hand tracking. For instance, an object may emit a burst of particles when grabbed, this adds juicy elements to the object and an additional feature to the object for players to remember them. This was achieved by configuring objects with the XR Grab Interactable component, which enabled physical manipulation, and attaching scripts to trigger feedback effect using Unity’s Particle System. The learning process involved setting up the XR Rig to track player movements, ensuring that interactions were intuitive and responsive. Iterative testing refined these objects to balance feedback intensity, preventing sensory overload and potential cyber sickness while maintaining immersion. This foundational work was crucial, as it established a robust framework for creating interactive elements that aligned with the project’s goals.

5.2 User Interface

The user interface was a critical component in the development of the VR juiciness artifact, designed to ensure an immersive and intuitive player experience. Unity’s UI system was employed to create 3D UI elements integrated into the virtual environment, enabling natural interaction through VR hand controllers. The 3D canvas was made grabbable using Unity’s XR Interaction Toolkit, allowing players to reposition them for optimal viewing, enhancing usability and accessibility. Text components were implemented with a billboard effect, rotating to the player regardless of their orientation.

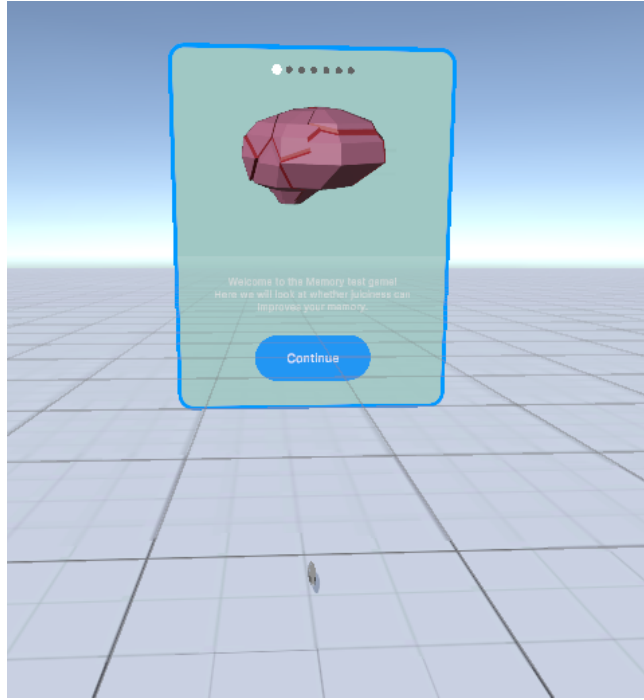


Fig. 7. The 3D UI used to inform participants about the memory game mechanics

To facilitate debugging Unity’s UI Toolkit was utilised for on-screen monitoring, providing a real-time view of UI states while participants engaged with the mini-games, this was not visible within the game itself but showed player scores and progress in the Unity play window to view scores without interrupting play. Initially, UI Toolkit was expected that Unity could support UI Toolkit with 2D UI within VR space, but this was found to be un-usable and an example of the pitfalls of testing VR using the XR Toolkit VR Rig for testing without a headset, as this was visible within unity but not when wearing the equipment. Instead, the focus was shifted to 3D UI, which connected to hand controllers for immersive interaction. The UI design process had a base design around the Unity XR sample scene, which provided a practical template and learning resource for creating effective VR interfaces as well as useful resources for user affordance and user experience. The result was a fully implemented 2D UI for data collection by the researcher, ensuring this development time on unusable 2D UI did not go to waste, as well as engaging 3D UI which added to the immersive experience of the VR world created.

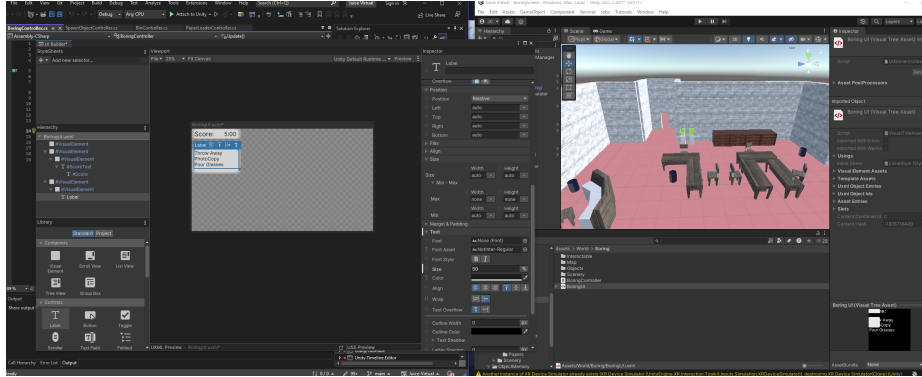


Fig. 8. UI Toolkit implementation within the boring mini-game

Beside the pitfall of using 2D UI Toolkit and it's incompatibility with VR headsets, 3D UI was driven by the need to create an immersive experience that aligned with VR's object focused game design principles. Unlike traditional 2D games, VR requires interfaces that players can interact with physically to maintain a sense of presence.

Unity's XR sample scene were a key resource, providing practical examples for VR UI design. It included templates for world space canvases, grabbable objects connected to UI elements, and interaction elements like buttons responding to hand motion and raycasting from the VR equipment. Additional opportunities for juiciness also arose through the lessons learnt from this scene as the controller would buzz when hovering over selectable elements and this taught improved game feel and juicy within the UI, not just the objects themselves.

5.3 Memory Game

The memory mini-game was developed to investigate the influence of juiciness on a players ability to recall features of objects, using juiciness as an additional feature attached to an object for players to remember. This mini-game challenges players to memorise the locations and features of items as well as to identify the kinds of items they were, after an amount of time, players must then find an item asked for, one at a time, as quickly as possible, once all items have been collected the game ends and a time is provided to the player as their score. By comparing a juicy version with a non-juicy version, the study aimed to quantify how juiciness impacts cognitive performance and engagement in VR.

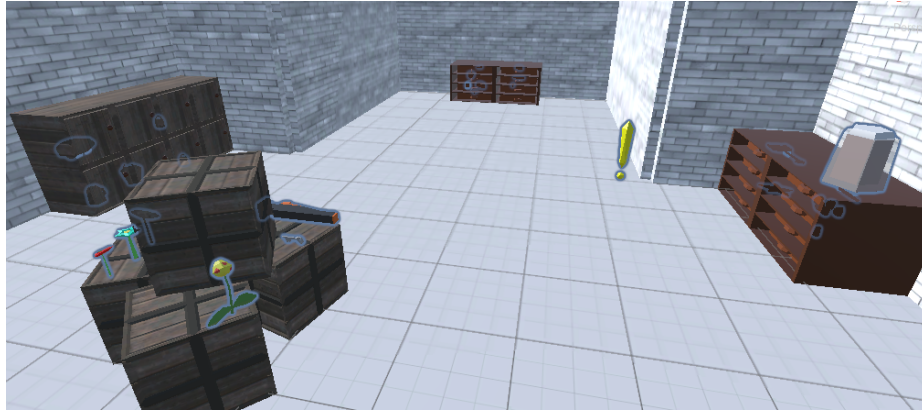


Fig. 9. The memory room with Juice Objects highlighted in blue

The code for the memory mini-game controller was within the 'ObjectU-IMemoryScript', this includes a countdown from five minutes and connecting this timer to the 3D canvas and 2D UI Toolkit UI within the scene, this timer is then preceded by a randomised object selection that takes a vector of 'JuiceObjects' that exist within the game, some objects could be destroyed as part of their juicy interactions and these were not included in the final randomised selection if this were the case.

```
2 references
public void RandomiseObject()
{
    bool found = false;
    NextItemTextPhysical.GetComponent<TextMeshProUGUI>().text = "Next Item:";
    int random = 0;
    Transform choice = null;
    int count = 0;
    itemsFound++;
    if (itemsFound == 5)
    {
        return;
    }
    while (!found)
    {
        random = Random.Range(0, AllJuice.Length);
        choice = AllJuice[random].GetComponent<Transform>();
        if (!alreadyFoundObjects.Contains(choice.name) && choice.gameObject.activeSelf) || count == 100
        {
            found = true;
        }
        count++;
    }
    ItemText.text = AllJuice[random].name;
    ItemTextPhysical.GetComponent<TextMeshProUGUI>().text = ItemText.text;
    CurrentJuice = AllJuice[random].GetComponent<JuiceObject>().name;
    ItemTextPhysical.GetComponent<TextMeshProUGUI>().text = CurrentJuice;
    alreadyFoundObjects.Add(choice.name);
}
```

Fig. 10. This code takes existing juice objects within the scene and asks the user to find one

The count down routine was also mimicked with a count up routine which worked in the opposite way to calculate the time taken by users and therefore their overall score for quantitative results. In this way the integration of the MemoryScript and the UI was interlinked, as the information about the mini-game needed to be shown to participants giving necessary information about the game goals and current tasks.

```

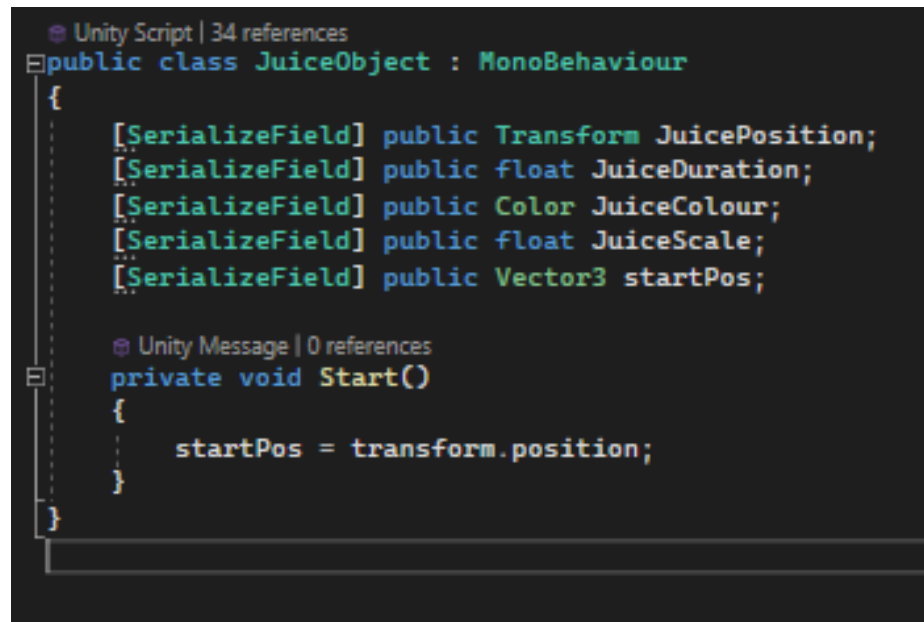
1 reference
IEnumerator CountUpRoutine()
{
    int TimePassed = 0;
    while (true)
    {
        if(itemsFound == 5)
        {
            NextItemText.text = "";
            NextItemTextPhysical.GetComponent<TextMeshProUGUI>().text = NextItemText.text;
            ItemText.text = "Result";
            ItemTextPhysical.GetComponent<TextMeshProUGUI>().text = ItemText.text;
            break;
        }
        yield return new WaitForSeconds(1);
        TimePassed++;
        seconds = TimePassed % 60;
        if (seconds == 0 || seconds == -1)
        {
            Minutes++;
        }
        if (seconds > 9)
        {
            TimeCount.text = Minutes.ToString() + ":" + seconds.ToString();
            TimeCountPhysical.GetComponent<TextMeshProUGUI>().text = TimeCount.text;
        }
        else
        {
            TimeCount.text = Minutes.ToString() + ":0" + seconds.ToString();
            TimeCountPhysical.GetComponent<TextMeshProUGUI>().text = TimeCount.text;
        }
    }
}

```

Fig. 11. This code controls a timer while turning that time into a visual clocks for users in VR

5.4 Juiciness Implementation

The game was set in a virtual room populated with a variety of items, such as a book, a talking plant and flowers, these objects did not follow a theme of any kind. Objects were configured with the XR Grab Interactable component, enabling players to pick up and manipulate items using VR controllers, allowing natural and intuitive interactions. Each item was given custom C script components relating the item but deriving from a 'Juice Object' Parent class, that held all information necessary to create juicy effects relevant to each object and that were serialised to be editable within the Unity editor. Within the memory mini-game, this class was used to determine which objects were fit to be requested by the item collection section of the test. All items within other mini-games also had this 'Juice Object' parent class in later games as well given the close integration between the setting of juicy objects and the execution of juicy effects.

A screenshot of the Unity Inspector window showing a C# script named 'JuiceObject' which inherits from 'MonoBehaviour'. The script has five public fields, each with a '[SerializeField]' attribute: 'JuicePosition' of type 'Transform', 'JuiceDuration' of type 'float', 'JuiceColour' of type 'Color', 'JuiceScale' of type 'float', and 'startPos' of type 'Vector3'. Below these fields is a private 'Start()' method that sets 'startPos' to 'transform.position'. The script has 34 references and 0 references to the 'Start()' method.

```
Unity Script | 34 references
public class JuiceObject : MonoBehaviour
{
    [SerializeField] public Transform JuicePosition;
    [SerializeField] public float JuiceDuration;
    [SerializeField] public Color JuiceColour;
    [SerializeField] public float JuiceScale;
    [SerializeField] public Vector3 startPos;

    Unity Message | 0 references
    private void Start()
    {
        startPos = transform.position;
    }
}
```

Fig. 12. This is the Juice Object parent class

Developed within the memory mini-game was the 'Juice Singleton' which was accessible from any object within the scene, this took the 'juice object' class and used the parameters set to determine the kind of juice an object released. This allowed quick creation of new juicy objects through calling the 'Juice' function as it served as a centralised and re-usable centre for the creation of new juice items.

```

21 references
public void Juice(JuiceObject juiceObject, JuiceTypes type)
{
    if (JuiceConstant.JuiceActive)
    {
        Transform juice = GetJuiceFromEnum(type);
        for (int i = 0; i < juice.childCount; i++)
        {
            if (!juice.GetChild(i).gameObject.activeSelf)
            {
                if (juice == this.ConstantStreamT)
                {
                    ConstantStreams.Add(juice.GetChild(i));
                    ConstantObjects.Add(juiceObject);
                }

                juice.GetChild(i).position = juiceObject.JuicePosition.position;
                var size = juice.GetChild(i).GetComponent<ParticleSystem>().shape;
                var main = juice.GetChild(i).GetComponent<ParticleSystem>().main;
                var colorOverLifetime = juice.GetChild(i).GetComponent<ParticleSystem>().colorOverLifetime;
                colorOverLifetime.enabled = false;
                main.startSize = juiceObject.JuiceScale;
                size.scale = new Vector3(juiceObject.JuiceScale, juiceObject.JuiceScale, juiceObject.JuiceScale);
                juice.GetChild(i).gameObject.SetActive(true);
                ColourParticles(juice.GetChild(i).GetComponent<ParticleSystem>(), juiceObject.JuiceColour);
                juice.GetChild(i).GetComponent<ParticleSystem>().Play();
                StartCoroutine(StopRoutine(juiceObject, juice.GetChild(i)));
                break;
            }
        }
    }
}

```

Fig. 13. The Juice function takes a juice object and handles the creation of juicy effects using the juice objects settings

The Juice function as seen in figure. 12, could be called from any object, however the 'JuiceConstance.JuiceActive' boolean defined whether juice would appear when asked, this implementation allowed the non juicy and juicy mini-games to be the same scene, speeding up development time and saving memory in game storage, this ensured the experiences of both mini-game versions were similar experiences barring the difference in juiciness.

Juicy effects were implemented and called upon by the 'Juice Singleton', these were related to an enum within a switch which allowed the 'Juice' function to call multiple kinds of juicy effects without the need for additional code, this meant changing a drop-down option within the Unity editor would change the effects and drastically increased development speed. These effects were stored within a container which used object pooling for optimisation purposes, this meant juicy effects would be stored within the game prior to their use and looped through to prevent the creation and destruction of objects, saving on memory and potential memory leaks.



Fig. 14. The juice container with all the juicy effects

5.5 Boring Game

The Boring mini-game was developed to investigate whether juiciness can increase player motivation in performing repetitive tasks in a mundane environment. The game was set in an office and included three tasks: photocopying documents, filling cups with water, and throwing objects in the bin, each task awards points upon completion. The player was able to decide when to exit the game. This design enabled the study to assess whether the addition of juicy feedback motives players to engage longer and achieve higher scores.

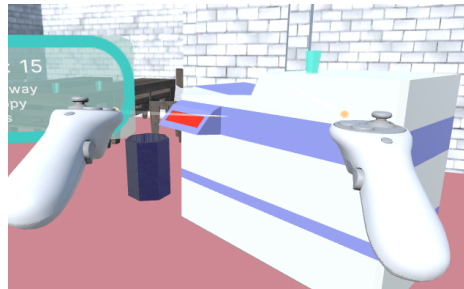


Fig. 15. This printer is indicating it contains no paper with a red screen

The printer itself was a 'Juice Object' like the objects created for the memory game, this meant easy access to the Juice Singleton 'Juice' function using itself as an input, the printer could take an infinite amount of paper, using a co-routine to print paper one at a time, repeating the juicy splash for every instance of score increase, accentuating the player feedback. A panel on the printer showed whether it was filled with paper, printing or empty, after printing paper.

```

1 reference
void PlacePaper()
{
    for (int i = 0; i < paperContainer.childCount; i++)
    {
        if (!paperContainer.GetChild(i).gameObject.activeSelf)
        {
            paperContainer.GetChild(i).transform.position = transform.GetChild(0).transform.position;
            Quaternion newRot = transform.GetChild(0).transform.rotation;
            paperContainer.GetChild(i).rotation = newRot;
            paperContainer.GetChild(i).gameObject.SetActive(true);
            paperCount--;
            BoringController.Instance.GetScore(5);
            return;
        }
    }
}

1 reference
IEnumerator PrintPapersRoutine()
{
    int currentCount = paperCount;
    for (int i = 0; i < currentCount; i++)
    {
        yield return new WaitForSeconds(1);
        JuiceSingleton.Instance.Juice(this, JuiceSingleton.JuiceTypes.SmallSplash);
        PlacePaper();
    }
    if (paperCount > 0)
    {
        NoPaper(false);
    }
    else
    {
        NoPaper(true);
    }
}

```

Fig. 16. These functions show the co-routine placing paper and the juicy feedback rewarding the player

Throwing away items is an interactable mundane task which also re-fills object containers within the scene, objects such as paper and plastic cups can be spawned from spawn points as well as printed paper from the printer, object pooling utilises all required memory at run time and prevents performance issues later, however players may run out of objects to spawn, therefore this bin mechanic allows the cyclic of objects from the game back into the object containers, the object collides with the bin and a juicy effect is spawned. This is an example of a game design feature that attempts to counteract the short comings of the game design architecture, the result being players during testing did not run out of objects or performance issues, despite large numbers of papers and cups being spawned.

```

@ Unity Message | 0 references
private void OnTriggerEnter(Collider other)
{
    other.gameObject.SetActive(false);
    JuiceSingleton.Instance.Juice(this, JuiceSingleton.JuiceTypes.SquareSplash);
    BoringController.Instance.GetScore(15);
}

```

Fig. 17. Objects collide with the bin, disappear from the player view, and reward the player

5.6 Time Trail Game

The time trial mini-game was developed to assess whether juiciness can improve player performance and competence in a skill, timed environment. The game required players to navigate a dynamic obstacle course while dodging cannons and shooting targets while completing the course in as fast a time as possible. The game began with a UI displaying instructions for the course, with a timer starting once players finished reading. Players needed to utilise VR teleportation movement to traverse the course and grab the button at the end to stop a timer and get an overall score.

The teleportation mechanic was available in all mini-games, the intention was to reduce cyber-sickness as discussed in Prithul et al. (2021). Players were able to point their controller and teleport to any position, which removed any need to walk, a common movement type in any games which can cause increased player dizziness and nausea in VR. This teleportation movement type allowed a new interesting way to develop the obstacle course, as such players were able to move as fast as their teleportation allowed instead of the slow movement of normal walking, changing the pacing and overall feel of the game. Overall this movement type was better suited to show player engagement and the difference between juicy and non juicy implementation on player competence.

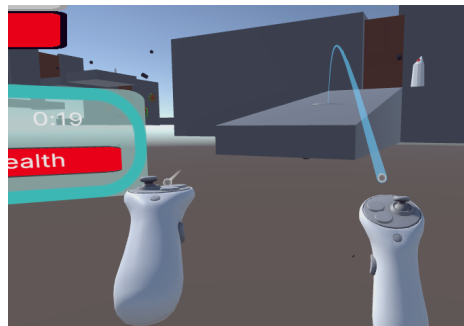


Fig. 18. The player is using the teleportation feature to navigate a gap impassable by walking

6 Chapter 6: Results & Discussion

6.1 PXI Results

The PXI questionnaire was developed by Vanden Abeele et al. (2020) to provide a better understanding of how game design choices impact player actions, in this way, the PXI was used to determine whether juiciness had any meaningful interaction with the overall player experience. Some participants within this study experienced only the juicy mini-games or the non-juicy version, however for most participants, they experienced at least one mini-game that was the opposite to the majority of their juiciness choices. Participants were told which mini-game juiciness choice they should play and no participant played both versions of a single mini-game. Care was taken to make sure an equal number of participants played each version of a mini-game. The PXI is designed to measure specific constructs of player experience, and each construct is measured with three similar items.

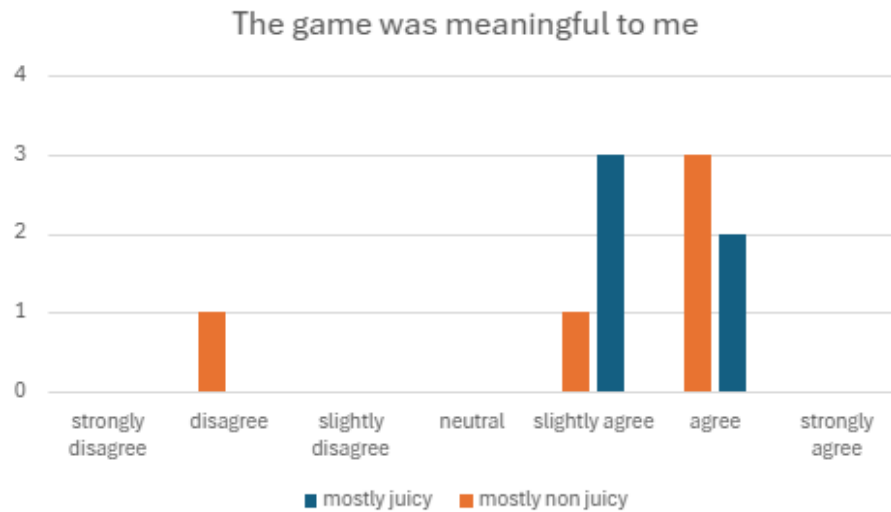


Fig. 19. This graph shows the results of whether participants found the game meaningful

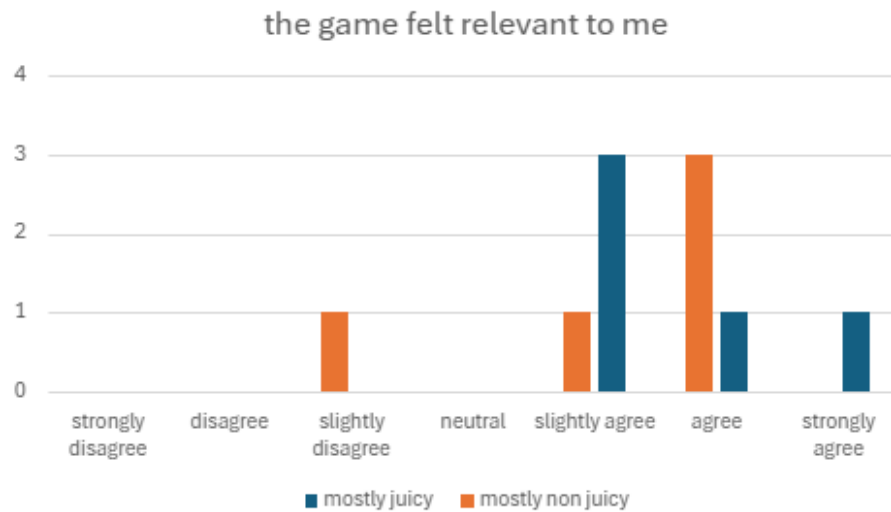


Fig. 20. This graph shows whether participants found the game relevant to themselves

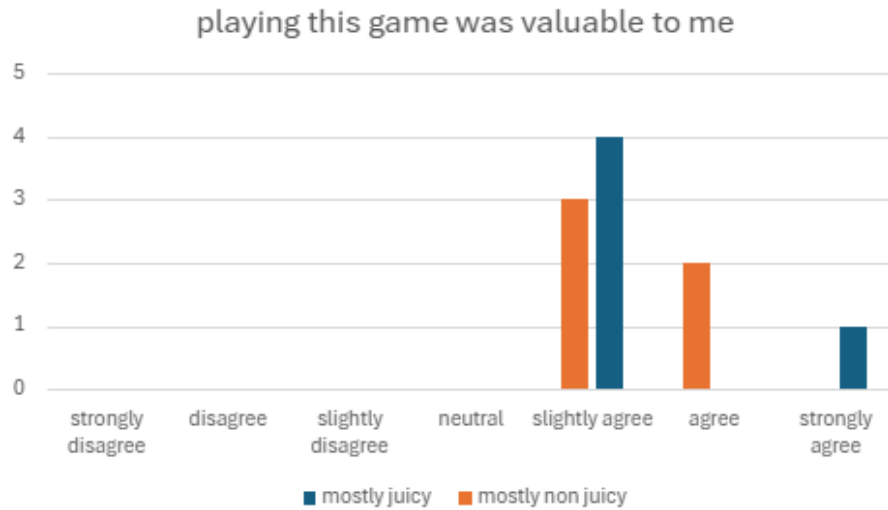


Fig. 21. This graph shows whether participants found the game valuable to themselves

The graphs seen within figures 19, 20 and 21 all refer to the extent to which players found meaning within the game itself, generally it was found this meaning had a negative skew, leaning towards agreement, 90% of participants found the game to be meaningful to them through all three questions, including meaning, relevancy and value. The results of this show a general interest in virtual reality and the novelty they experienced. Generally juicy and non juicy participants agreed on these values, however one non-juicy participant disagreed on meaningfulness and relevancy while still viewing the game as valuable.

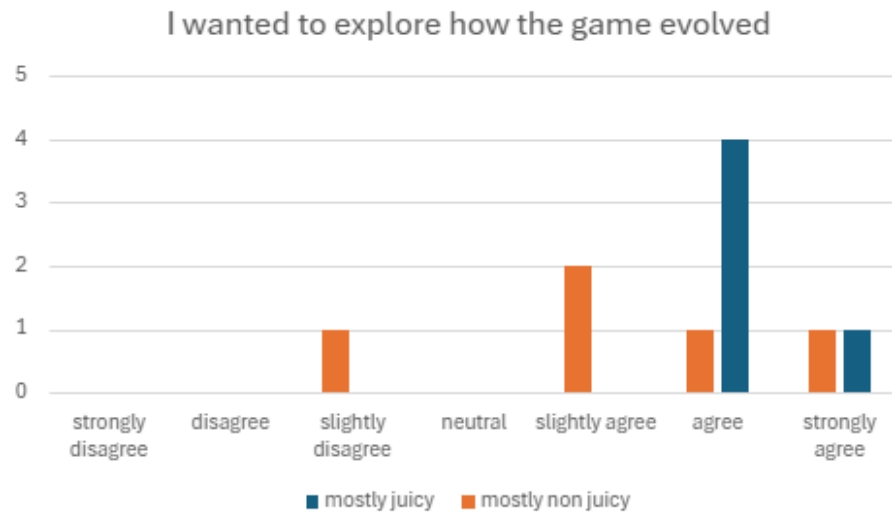


Fig. 22. This graph shows the results of whether participants were interesting how the game evolved

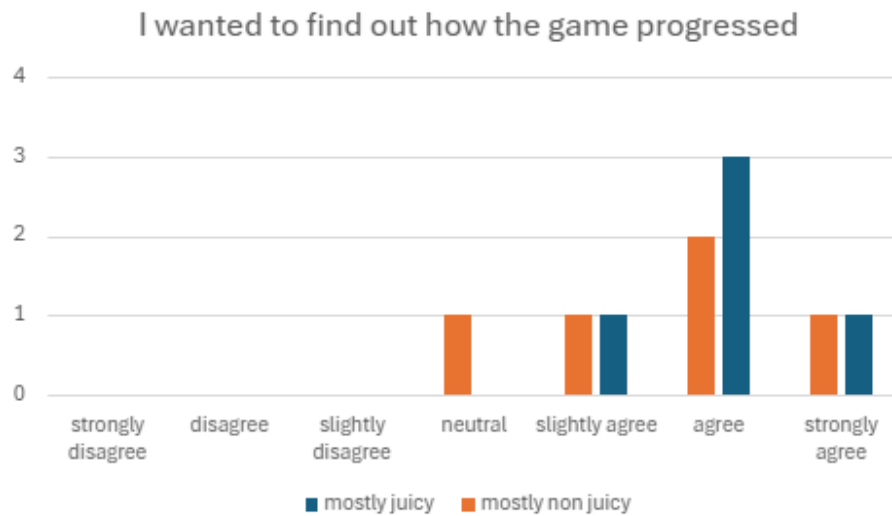


Fig. 23. This graph shows whether participants were interested in how the game progressed

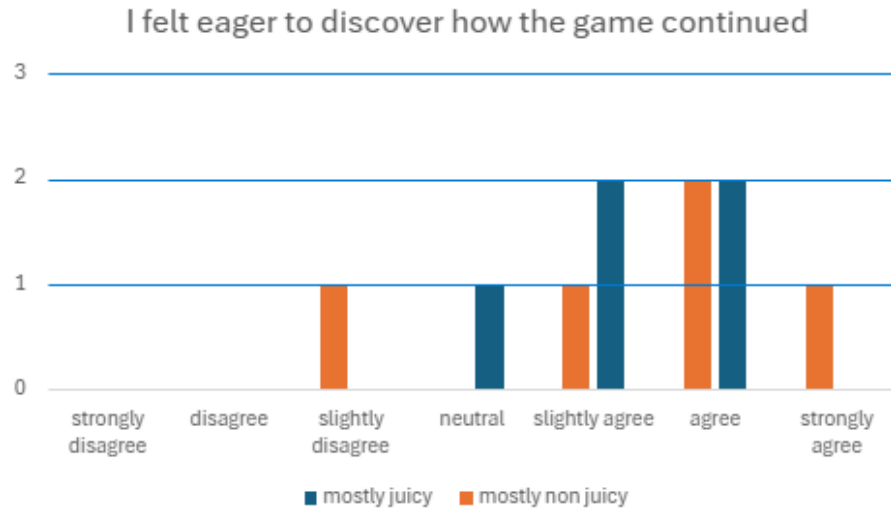


Fig. 24. This graph shows the results of whether participants were interested in how the game continued

The graphs within figures 22, 23 and 24 discuss the constructs within the PXI regarding curiosity, overall curiosity had a negative skew towards agreement, however there was a slight different between juicy and non-juicy participants, while both groups tended towards agreement, non-juicy participants had a wider range between slight disagreement and agreement while juicy participants slightly agreed or strongly agreed. For discovery of how the game continued, seen in figure 24, a much larger range for non-juicy participants was found. Most notably, juicy participants agreed significantly more on wanting to see how the game evolved, seen in figure 22.

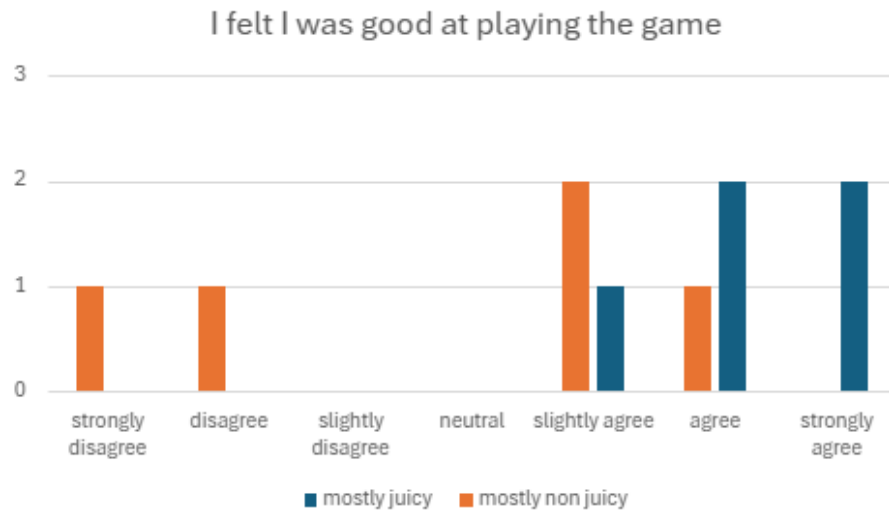


Fig. 25. This graph shows the results of whether participants thought the game felt good to play

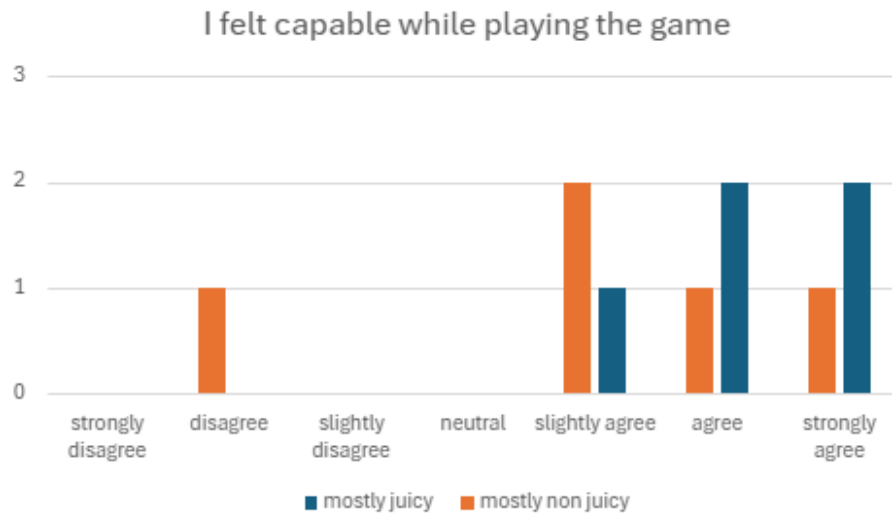


Fig. 26. This graph shows whether participants were felt capable playing the game

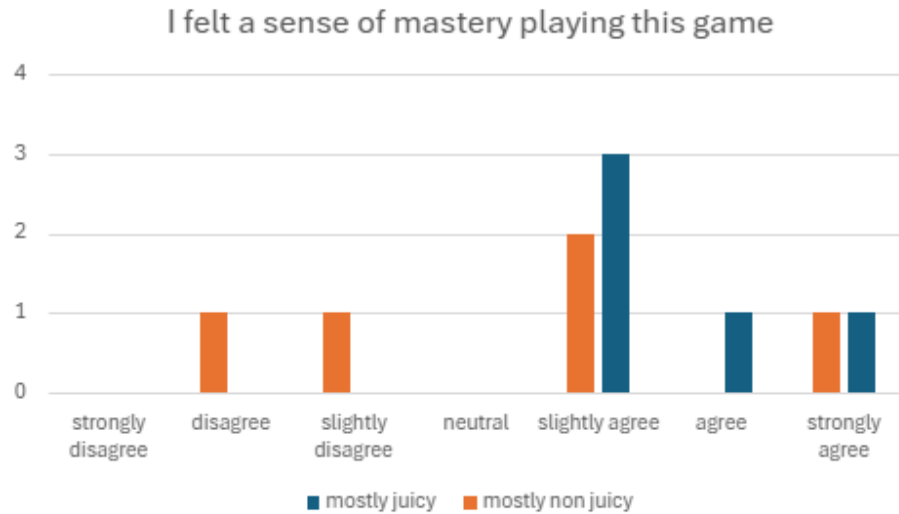


Fig. 27. This graph shows the results of whether participants felt mastery over the game

Graphs seen in figures 25, 26 and 27, show the difference in mastery that participants felt over the game, a notable difference was found from players of the non-juicy group to the juicy group, this construct within the PXI is very interesting as it pertains to competence, a subject explored within the time trial mini-game within the artefact. 60% of participants of non juicy groups felt good and a sense of mastery while playing. All participants of the generally juicy group felt a sense of mastery in all items within this PXI construct. These figures show a clear change in the effects juiciness had within the artefact created on perceived competence.

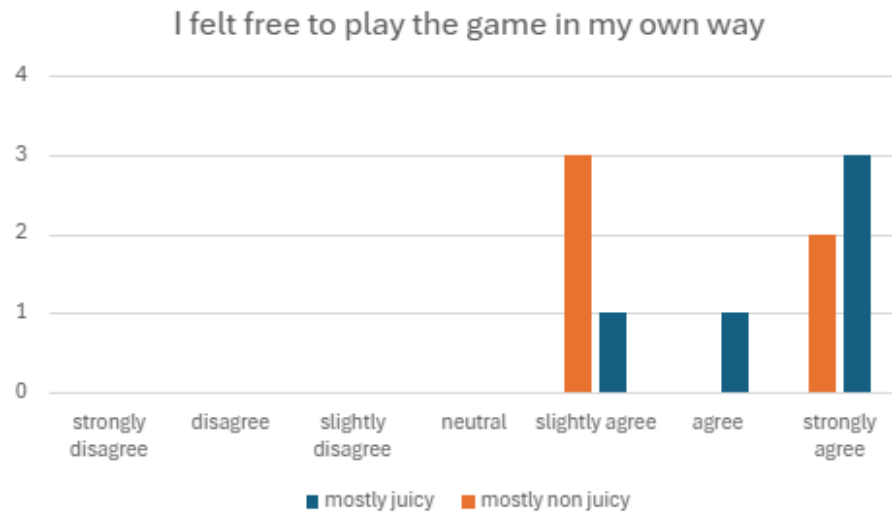


Fig. 28. This graph shows the results of whether participants felt free to play in their own way



Fig. 29. This graph shows whether participants felt they had a choice of how to play

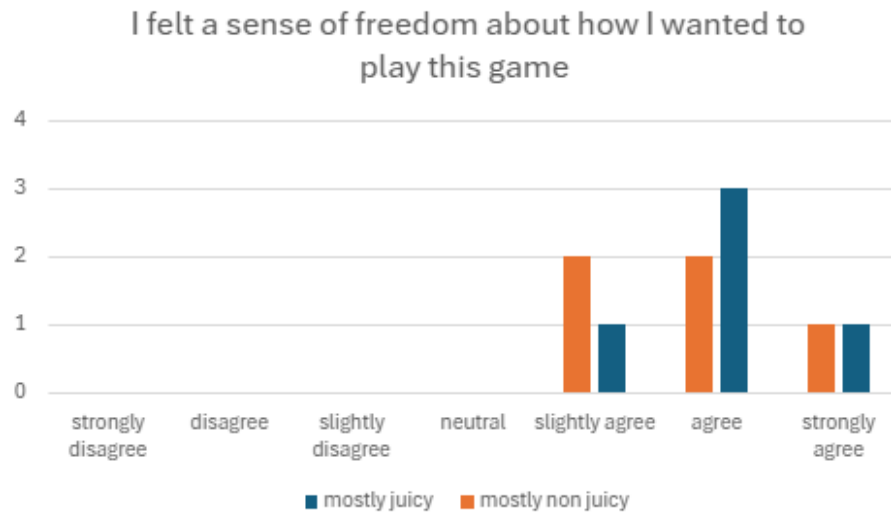


Fig. 30. This graph shows whether participants felt they had a sense of freedom while playing

Graphs within figures 28, 29 and 30 show results from the PXI questionnaire with regard to autonomy, it reveals the mostly juicy VR mini-games appeared to enhance the players sense of autonomy. The mostly juicy group reported higher agreement, particularly in the strongly agree category, suggesting juiciness may foster a greater sense of player agency. Although the mostly non juicy group also generally agreed with the question, they typically gave more neutral and moderate responses indicating a less compelling experience.

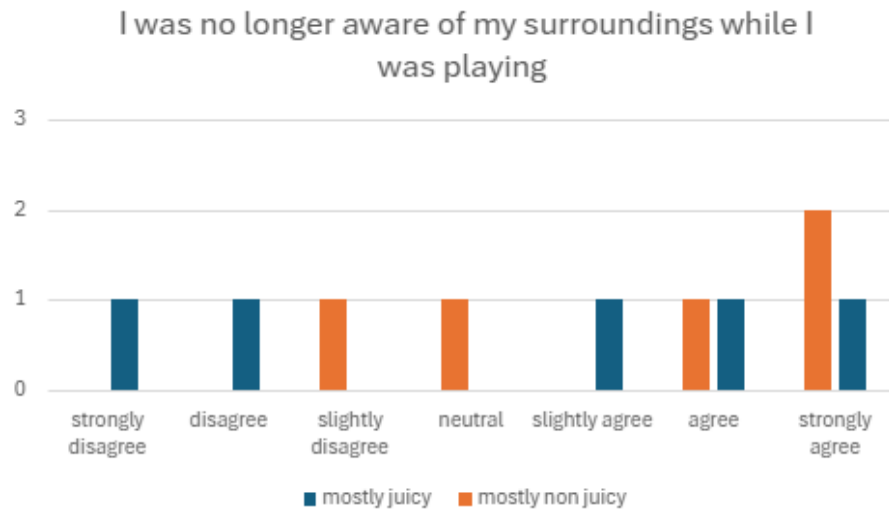


Fig. 31. This graph shows the results of whether participants were not aware of their surroundings while playing

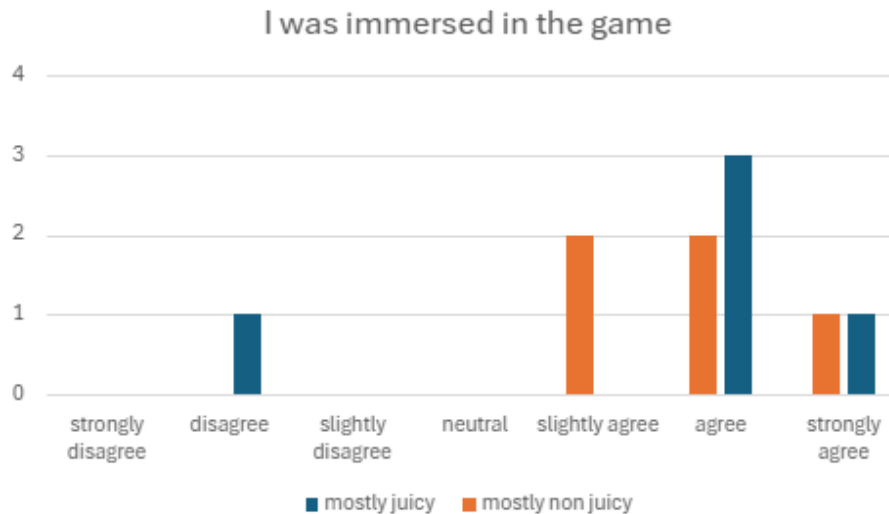


Fig. 32. This graph shows whether participants were immersed in the game

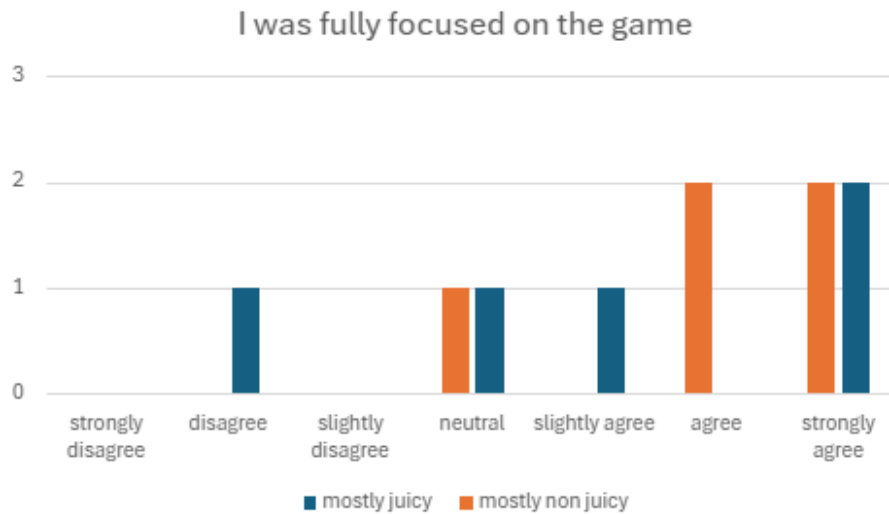


Fig. 33. This graph shows whether participants felt fully focused in the game

Figures 31, 32 and 33 refer to immersion, this is a key element of VR reality games, a wide spread of answers within figure 31 show mixed level for loss of self, with 60% of both participants groups agreeing, but with juicy participants disagreeing much stronger, this is followed by figures 32 and 33 where non-juicy participants were more likely to agree with immersion and game focus. 100% of non-juicy participants felt immersed in the game with 80% feeling focused compared to 60% positive responses from the juicy group. This could be related to the way juiciness was viewed by the player within VR, Juicy interaction may decrease player immersion by reminding the user they are in a game, juicy effects do not exist within the real world which virtual reality emulates more effectively than traditional games.

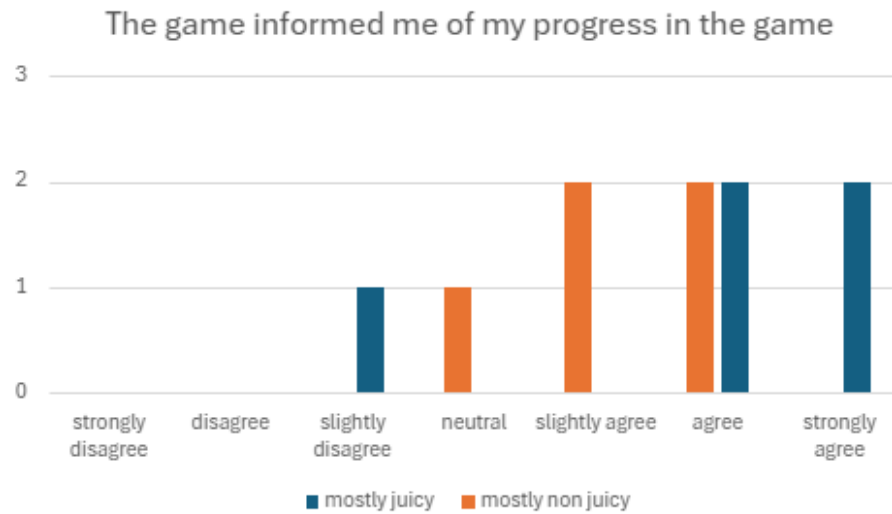


Fig. 34. This graph shows whether participants felt informed of their progress

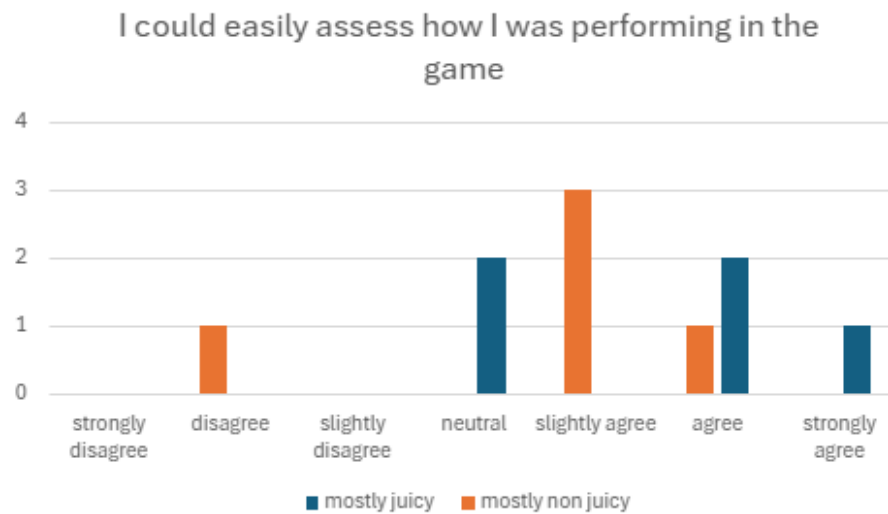


Fig. 35. This graph shows whether participants felt they could access their game performance



Fig. 36. This graph shows whether participants felt the game gave clear feedback on their progress

Figures 34, 35 and 36 refer to the Player Experience Index Construct of progress feedback, this functional consequence shows the extend to which players felt they were informed on their achievements. Throughout all three graphs the mostly-juicy group appeared to be more agreeable. Within all three questions, 7 non-juicy responses only slightly agreed with the question while 7 Juicy participants agreed, 4 responses from juicy participants strongly agreed to the question compared to one from the non-juicy group. It is clear the juiciness improved participants ability to understand progress.

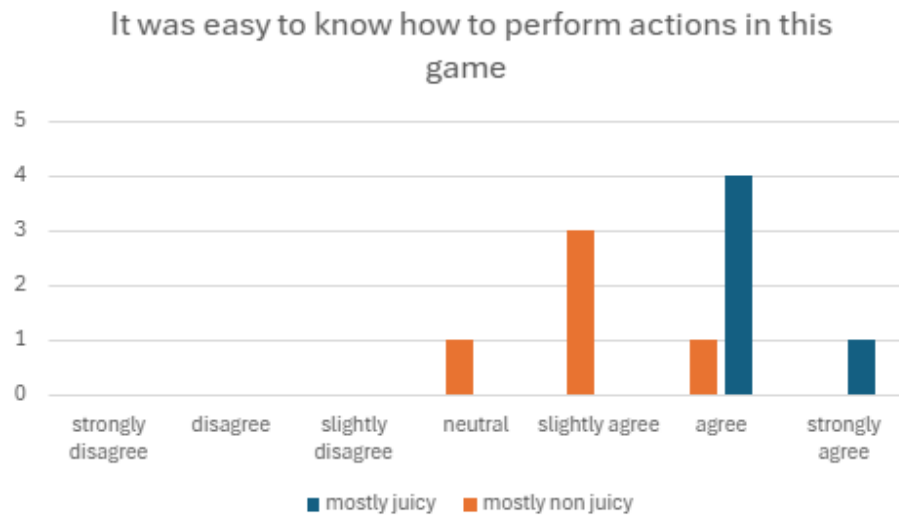


Fig. 37. This graph shows whether participants felt it was easy to know how to perform actions

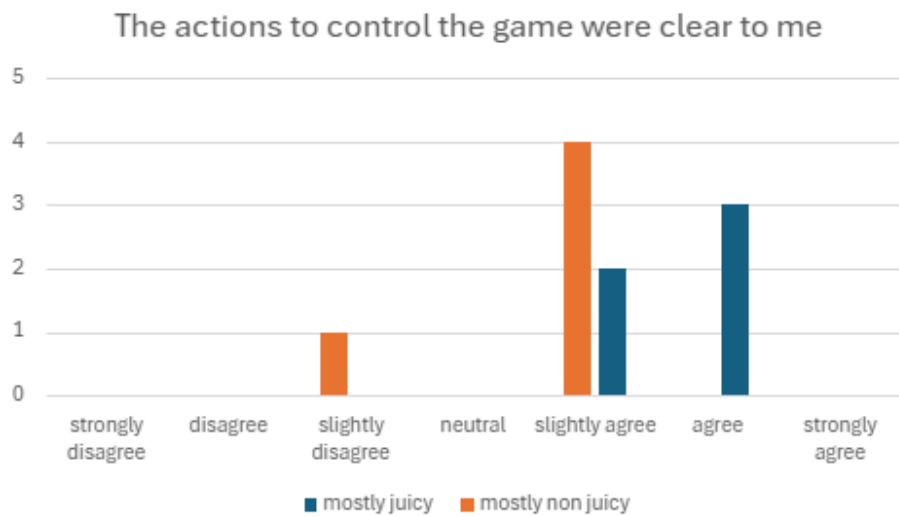


Fig. 38. This graph shows whether participants felt the actions to control the game were clear

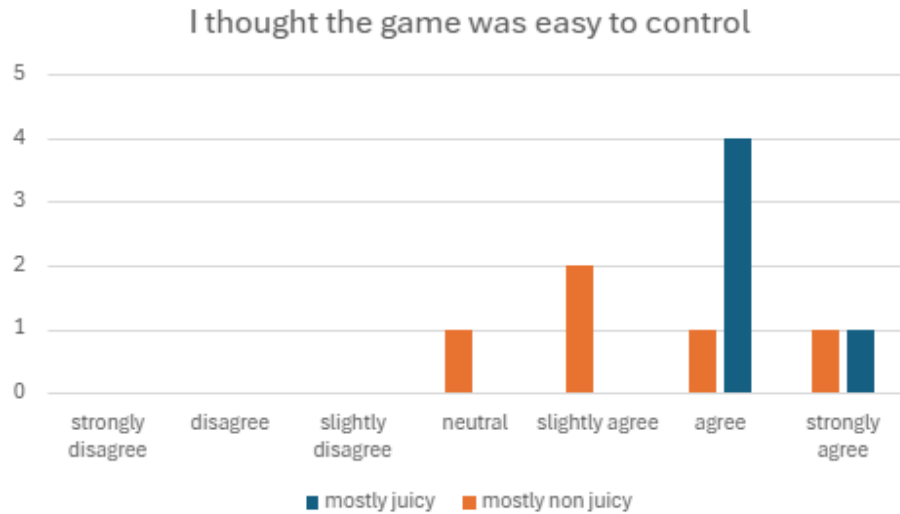


Fig. 39. This graph shows whether participants felt the game was easy to control

Within figures 37, 38 and 39, the results of the Ease of Control construct are given, it would appear there was an emphatic difference in the mostly-juicy participants viewed their ability to control the game. Out of all answers 80% of juicy participants agreed or strongly agreed with the question, on average participants agreed with the question with 11 responses, compared to non-juicy participants who gave 9 out of 15 responses being slightly agree. Juiciness appears to have increased the agreeableness of participants views on game controls by full level. It is probable from the results that juicy responses are particularly useful to improve players perceptions of game control and that this result carries over to be applicable and notably useful in virtual reality.

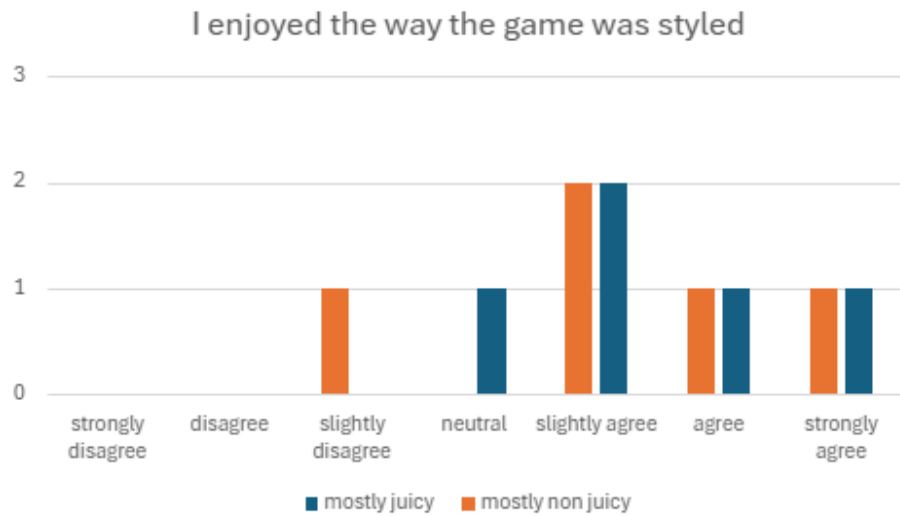


Fig. 40. This graph shows whether participants enjoyed the way the game was styled

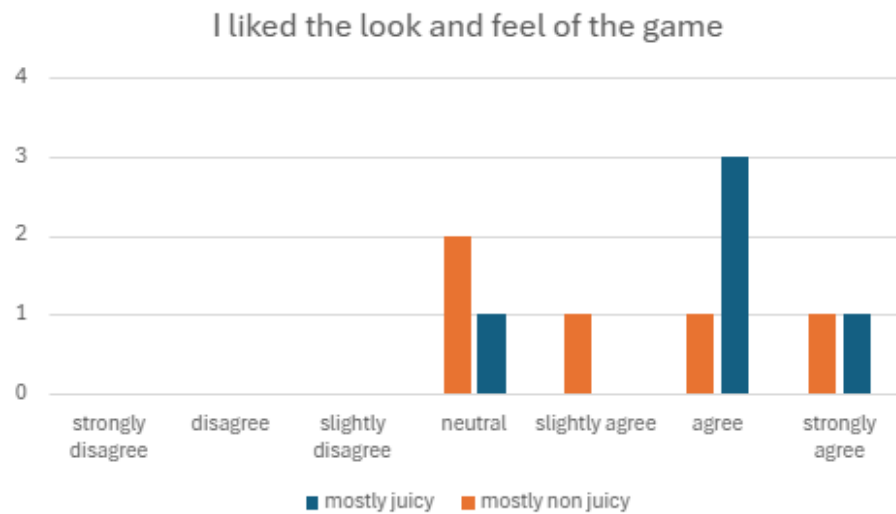


Fig. 41. This graph shows whether participants liked the look and feel of the game

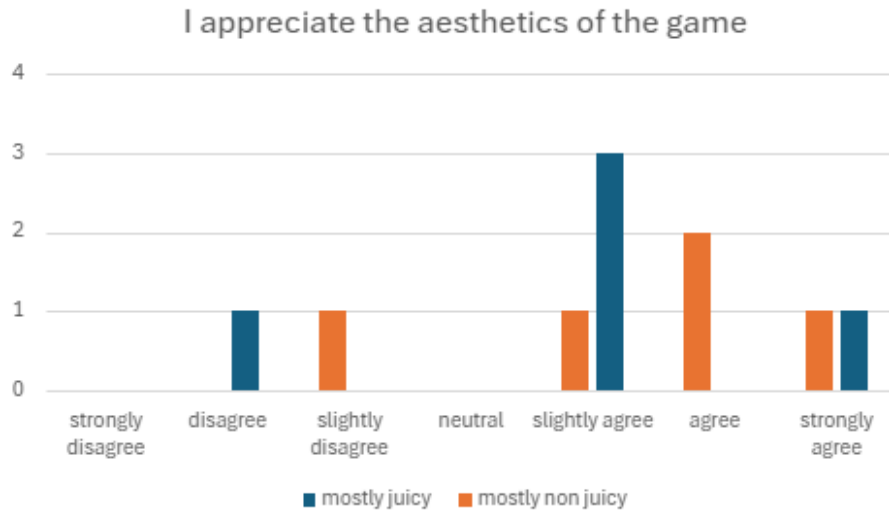


Fig. 42. This graph shows whether participants appreciated the aesthetics of the game

Overall figures 40, 41 and 42 show a positive response to the audiovisual appeal PXI construct within the game, Juiciness generally did not appear to change the style in most styles, however the mostly-juicy participants did respond notable more positively to the look and feel of the game, where 80% of participants responded agree or higher, and where 60% of non-juicy participants responded slightly agree or lower. This may be because juiciness added not to the style but to the game feel, this is usually the focus of juicy inclusions within games, as mentioned by Pichlmair & Johansen (2021) where juiciness is an important tool in designing game feel, in this way, juiciness is just one part of the overall aesthetics of a virtual reality game, but has a notable effect on feel.

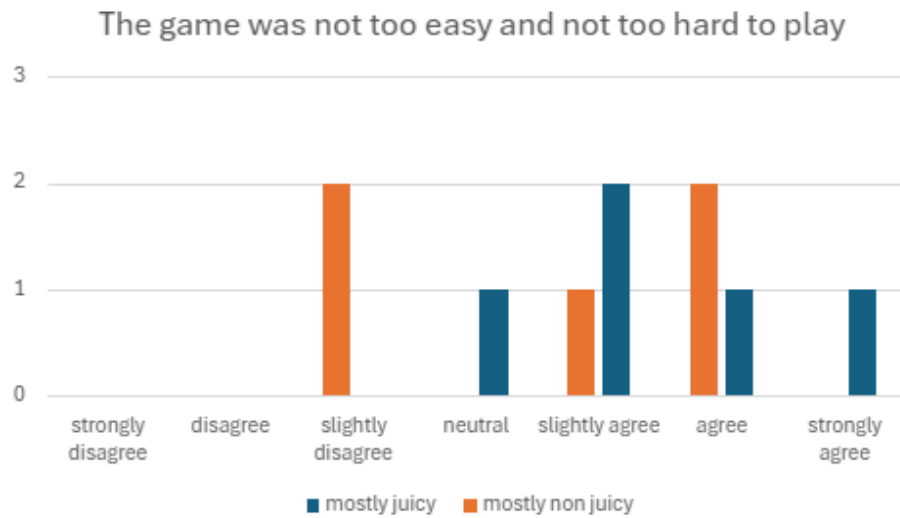


Fig. 43. This graph shows whether participants felt the game difficulty was the right level

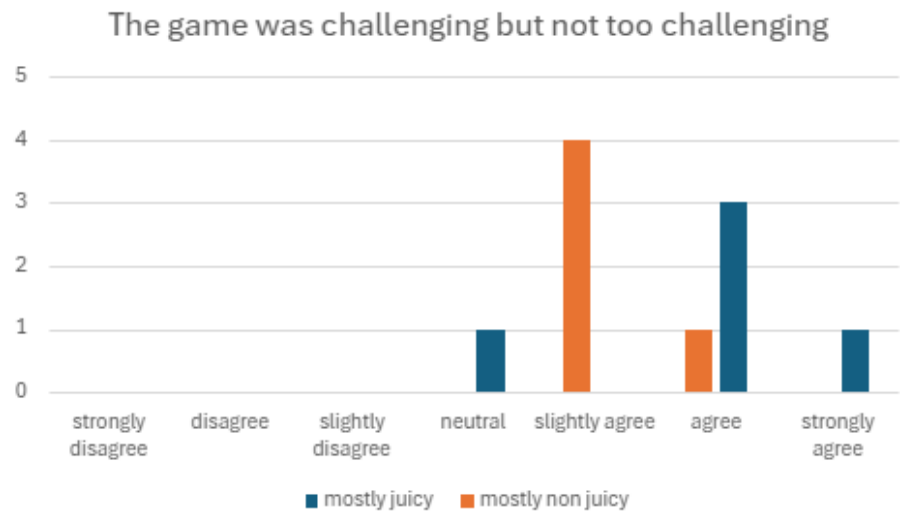


Fig. 44. This graph shows whether participants found the game slightly challenging

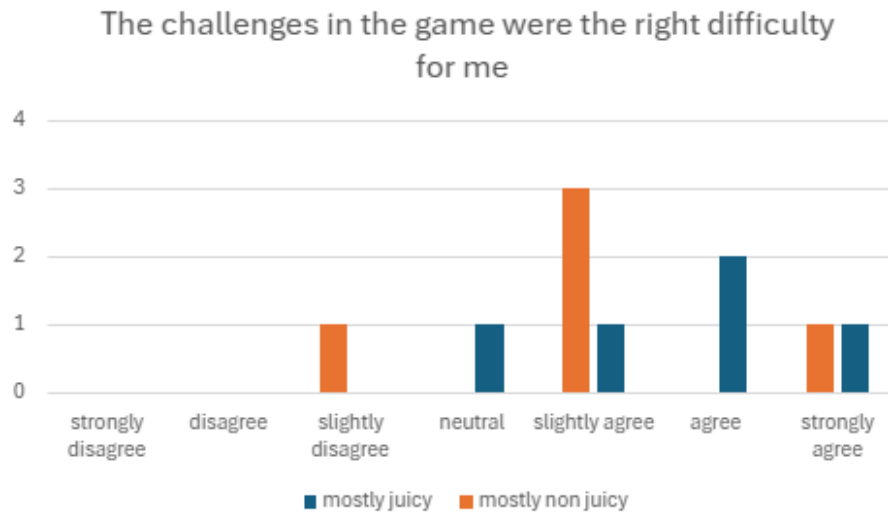


Fig. 45. This graph shows whether participants appreciated difficulty of the game

Figures 43, 44 and 45 discuss the challenge construct and the extent to which specific challenges in the game match the players skill level, the results show additional juiciness made the perception of challenge more acceptable, out of the three questions, both the non-juicy and juicy participants agreed with the question 12/15 times, however 9/15 juicy participants agreed or higher compared to 4/15 non-juicy participants, this show a strong drop off where the non-juicy group found the game challenge only slightly agreeable. Along with this, 3/15 responses slightly disagreed from the non-juicy participants compared to none in the three questions in the juicy group. It's difficult to argue that juiciness makes games tangibly easier, however using juiciness as a reward system can certainly decrease a sense of challenge and increase a self view on competence. Challenge can be related to competence, a concept tested within the time trial mini-game, these results show that regardless of the final quantitative results of the time trial mini-game provide, a players view of their overall ability is increased by juiciness.

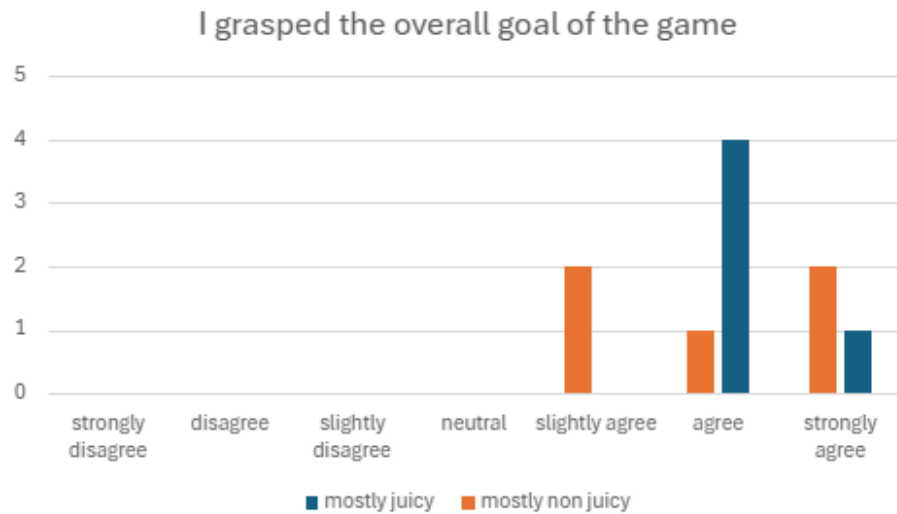


Fig. 46. This graph shows whether participants felt they grasped the goal of the game

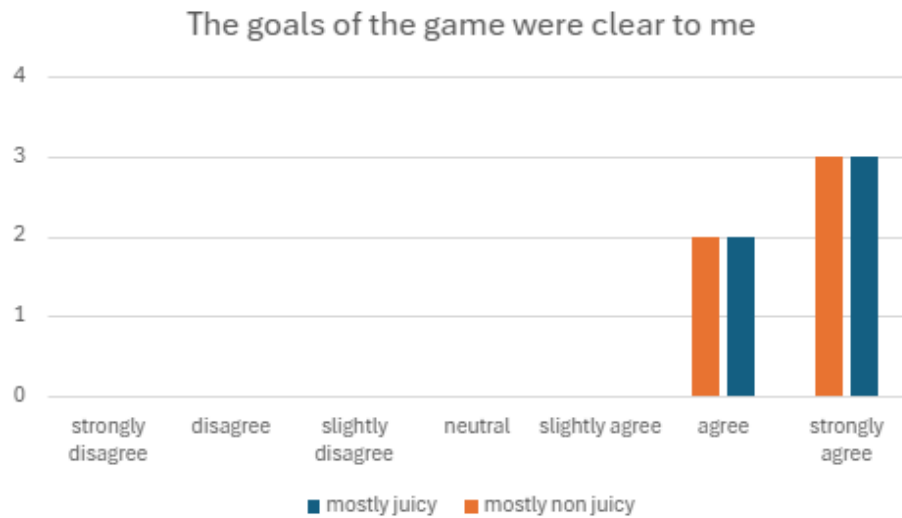


Fig. 47. This graph shows whether participants found the goal to be clear to them

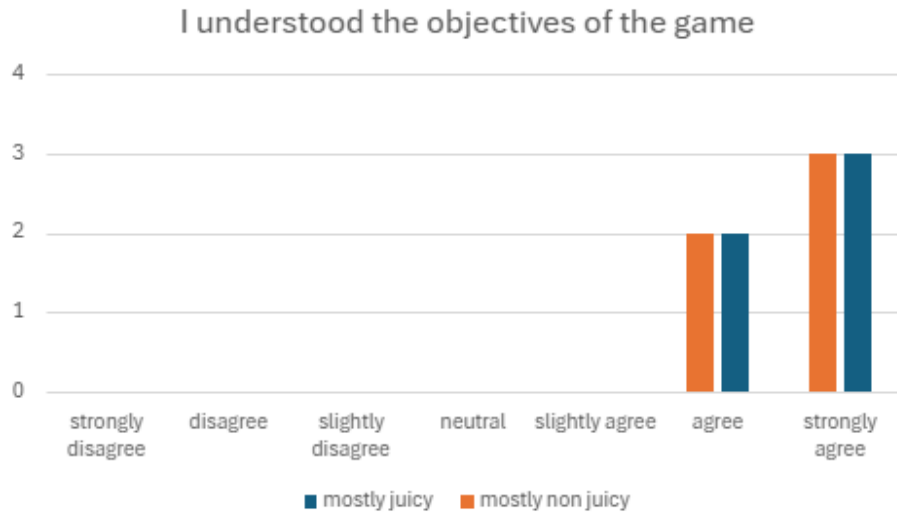


Fig. 48. This graph shows whether participants feel they understood the game objectives

Figures 46, 47 and 48 refer to the goals and rules construct of the PXI questionnaire, the results shown suggest juiciness didn't affect the participants ability to understand the mini-games, the greatest different between the two groups is shown within grasping the overall goal of the game, where all mostly-juicy participants agreed they understood while non-juicy participants were a little more mixed, despite this all groups agreed with these questions, interestingly, the results from figures 47 and 48 were exactly the same, this suggests juiciness had no bearing on a users perception of understanding goals and rules. This mostly makes sense as juiciness plays no role in describing a task to a user, and instead shows the effectiveness of the 3D UI implemented within the artefact, however more research could be performed regarding whether juicy responses reinforce a users belief of a games overall goals. Within figure 46, non-juicy participants agreed with the question on average with 2 answers slightly agreeing, and 2 agreeing fully, while no mostly-juicy participants answered slight agreement or less, perhaps juiciness reinforced the overall goals described at the start of each mini-game.

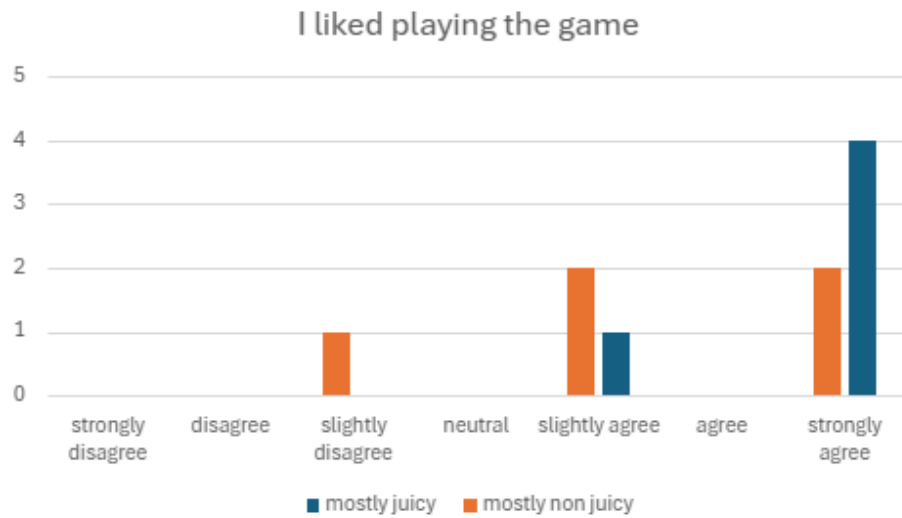


Fig. 49. This graph shows whether participants felt they grasped the goal of the game

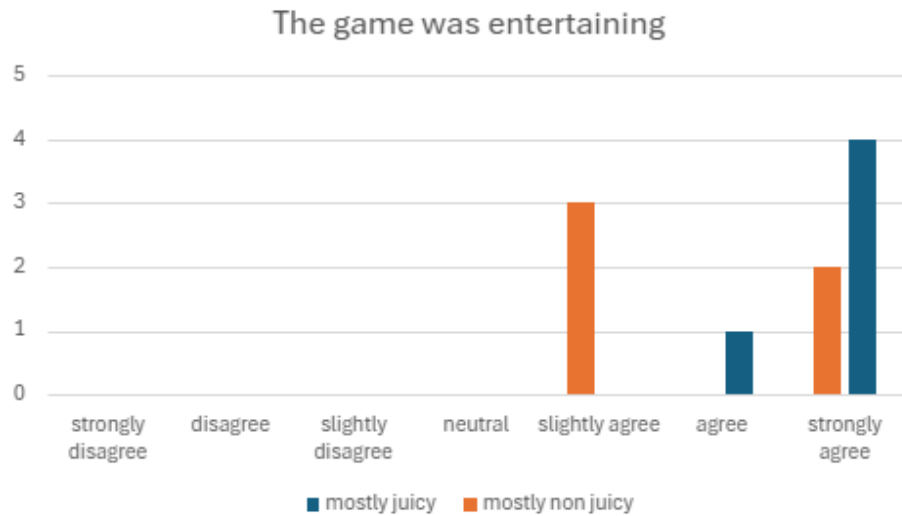


Fig. 50. This graph shows whether participants found the goal to be clear to them

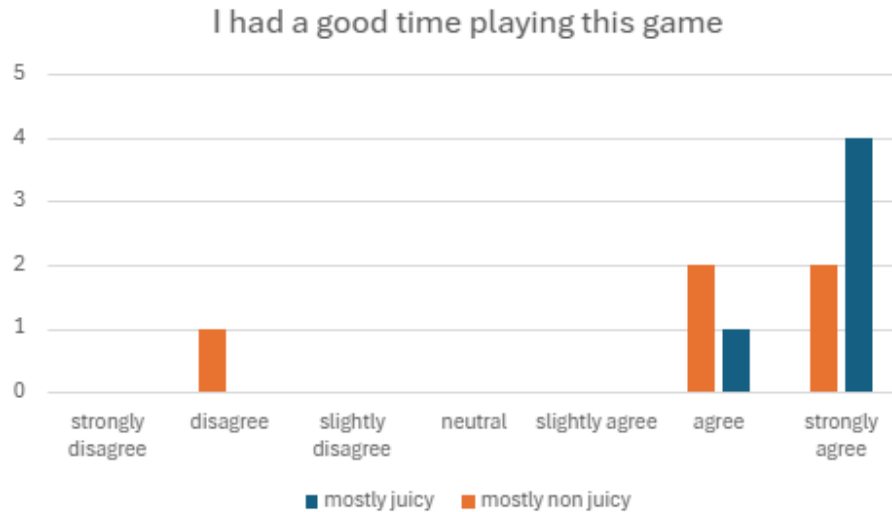


Fig. 51. This graph shows whether participants feel they understood the game objectives

The three graphs shown in figures 49, 50 and 51 relate to questions that were additional to the PXI and are generally related to overall player experience and enjoyment. Overall it is shown that 80% of the juicy group of participants strongly agreed with all three questions, this shows the overall increase is a positive experience through juiciness. Within the non-juicy group, it is possible a participant found the game to be a bad experience, this is surprising as the novel nature of virtual reality generally improves the opinion of virtual reality experiences, more could have been done to reduce cyber-sickness and improve the overall game design, however this member was not part of the mostly juicy group and since some participants didn't play any mini-games within the juicy variants, it's difficult to say whether adding additional opportunities for juiciness and the overall juicy quality could have changed this response. The juicy group had an overall better experience in their opinion with 87% of responses from the three questions returning as strongly agree., the most common response given from the non-juicy group was slightly agree with 33% of responses, showing a mixed view of the overall game without juiciness included.

6.2 Quantitative Results

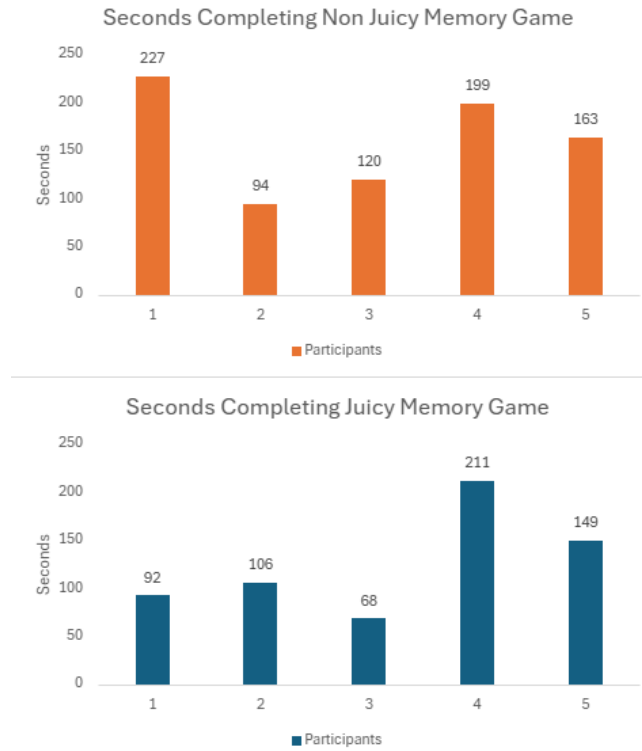


Fig. 52. These graphs show the time in seconds that participants took to complete the memory mini-game, orange is non juicy, while blue is juicy

The memory mini-game showed improved performance for the juicy participant group, the non-juicy median was 160.6 seconds compared to 125.2 seconds median for the juicy group, this means the non-juicy group were 28.3% slower on average, this shows a marked increase in memory caused by juicy feedback. The juicy scores experienced a the largest range being 143, compared to the non juicy group with 133 range, although these ranges are close, all non-juicy scores fit within an acceptable range. Participant 4 within the juicy group has a score about the upper quartile of expected data given it's median, discounting this value as an outlier, the range becomes 81, meaning juiciness could have a positive effect on the consistency of player outcomes, but this is difficult to conclude taking all results into consideration.

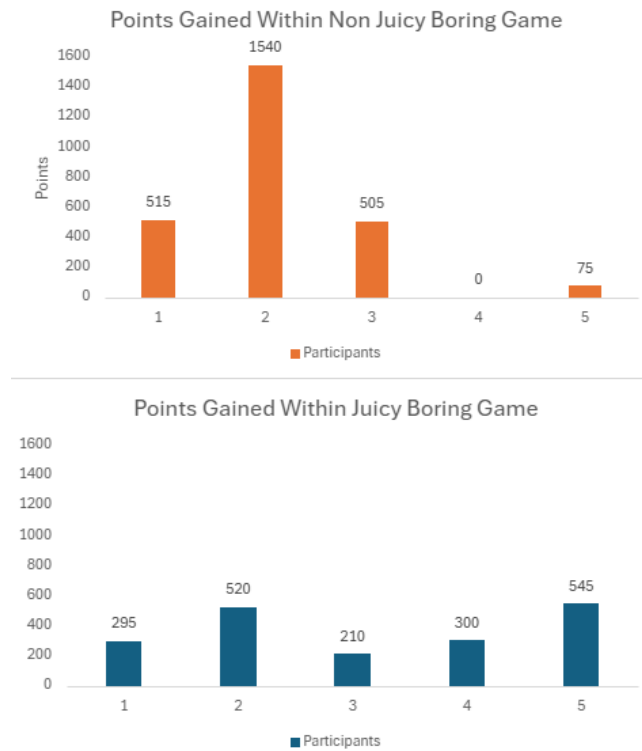


Fig. 53. These graphs show the score gained by participants by playing the boring mini-game, orange is non-juicy, while blue is juicy

The graphs shown above in figure 53 show the points gained by participants in the boring mini-game, overall the scores gained were relatively random, the median of the non-juicy scores was 658.75, one participant felt cyber-sickness to the point of not continuing the mini-game and did not gain a score. The median of the juicy game was 374, the scores gained were much more consistent but were lower overall. The outcome shows no quantitative data supports an increase in motivation from participants though this data is hard to compare given the large range of outcomes found within the non-juicy results, it's likely the most motivating factor of the mini-game was the novelty of using virtual reality itself.

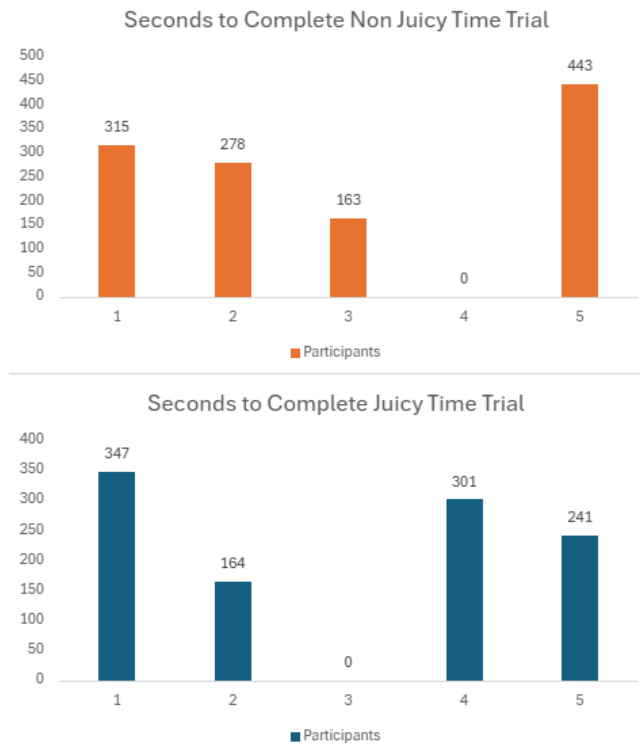


Fig. 54. These graphs show the seconds participants took to complete the time trial, where seconds were 0, the participant gave up after dying in game

The median time of the non-juicy participants as seen in figure 54 was 299.75 seconds, this is higher than the juicy participants with a median of 263.25, overall non-juicy participants performed 13% slower than juicy participants, this is a notable change in performance given participant 5 of the non-juicy group provided a significantly higher time than their other participants, despite this the score exists within the upper quartile of overall non-juicy scores and does not exist as an outlier.

Each group had a participant that did not finish the time trial, these chose to stop the mini-game after dying which would restart the course and timer. Three non-juicy participants had one death during the testing, one of which stopped the test, within the juicy participants, non died except for participant 3 who died twice before quitting, this may show an overall increase in focus within the juicy participants and a potential increase in motivation as well given the higher number of deaths before quitting.

6.3 Discussion

The outcomes of these tests provide an understanding of how juiciness impacts player experience, with regards to memory, motivation, and competence. The findings reveal both the strengths and limitations of juiciness in enhancing VR game-play, offering valuable insights for game design and potential applications in recreational, educational and instructional contexts.

Enhanced Player Experience and Performance It is evident that juiciness significantly influenced player experience, as demonstrated by the PXI results, and overall player performance in the memory and competence mini-games. The PXI data, encompassing constructs such as curiosity, mastery, autonomy, progress, ease of control, audiovisual appeal, and challenge, showed higher scores for the mostly-juicy group compared to the mostly non-juicy group. Quantitative performance metrics further corroborated these findings, with the juicy group achieving better times within the memory and competence related mini-games. This aligns with the study's aim to quantify whether VR amplifies juiciness effects, suggesting that enhanced feedback, creates a more engaging and effective game-play experience.

Neutral or Negative Effects on Certain Constructs Despite the overall positive impact, juiciness had no significant effect or, in some cases, a negative effect on the PXI constructs of goals, meaning, and immersion. For Instance, participants in the mostly juicy group did not report a stronger connection to the game's objectives or a deeper sense of purpose compared to the non-juicy group.

Juiciness also did not have a great effect on immersion, with both groups reporting comparable immersion levels. This could be due to the mini-games' focus on task based mechanics rather than a cohesive narrative which is crucial for immersion. These findings highlight the need for a balanced approach to VR game design where juiciness is complemented by elements such as storytelling or meaningful objectives to fully optimise player experience.

High Enjoyment Levels Across Both Conditions Participants across both the juicy and non-juicy groups reported high levels of enjoyment. The memory, motivation and competence mini-games were therefore found to be relatively well designed, ensuring participants remained invested throughout the study. This high enjoyment level validates the efficacy of the tests, as it indicates the results were not skewed by disengaged participants.

Improved Performance in Memory and Competence Quantitative results revealed a notable improvement in player performance within the memory and competence mini-games when juiciness was present. In the memory mini-game, participants collected items 28% slower within the non-juicy version. This

improvement could be due to enhanced feedback making interactions with objects more memorable. Similarly, in the competence mini-game, the non-juicy group performed 13% slower. These findings suggest that juiciness, combined with VR’s immersive qualities, positively affects cognitive and skill-based tasks in a meaningful way. The results open avenues for further research into how feedback design can optimise performance in VR training or educational applications.

Limited Effect on Motivation in the Boring Mini-Game The motivation mini-game, did not show a clear increase in motivation from juiciness. The variance in participants’ scores was high, making it challenging to draw definite conclusions, with the non-juicy group finding an overall higher average score before quitting. However, in the time trial mini-game, the juicy group exhibited a higher number of attempts before giving up if they died during the run. This suggests that juiciness may foster increased motivation in scenarios where immediate feedback aligns with performance goals. The mixed results highlight the context-dependent nature of juiciness’s impact on motivation, warranting further investigation into how feedback interacts with different game mechanics and player incentives.

Consistency and Leveling Effect of Juiciness Throughout the quantitative testing, the mostly juicy group exhibited more consistent scores and results compared to the non-juicy group, with a lower standard deviation in performance metrics across all mini-games. This consistency suggests that juiciness may provide a levelling effect, potentially making games more accessible to players of varying skill levels by enhancing engagement and focus. The levelling effect is an unexpected finding that may warrant further exploration because it may have implications for designing VR games that cater to diverse player abilities and preferences.

Implications for VR Game Design the study’s findings offer actionable insights for developers and researchers. The positive impact on memory and competence suggests that juiciness can be leveraged to enhance cognitive and skill-based tasks, making it a valuable tool for educational and training applications. However the neutral or negative effects on goals, meaning and immersion indicate that juiciness alone is insufficient for creating deeply meaningful or immersive experiences. Developers should complement juicy feedback with strong narratives and clear objectives to fully engage players.

7 Chapter 7: Conclusion

Overall this paper took steps to outline ways in which juiciness interacts with virtual reality, quantifying whether VR alters juiciness effects. The study employed three mini-games to test how enhanced feedback influenced player experience. Using the PXI questionnaire and quantitative performance data, the research demonstrated that VR's immersive environment amplifies the sensory impact of juiciness through visual cues. The findings align with the study's primary goal of exploring this interaction, providing a framework for understanding how Juiciness enhances a virtual reality experience.

Results show an overall increase in players' experience across most areas, though some PXI constructs were negatively affected. Quantitative data indicated that cognitive functions, including memory and competence, improved significantly with juiciness. However, motivation did not appear to be largely affected, as seen in the motivation mini-game where score variance was high and no trend emerged. This suggests while juiciness enhances task-specific performance and engagement in cognitive and skill-based contexts, its influence on sustained motivation may depend on game design and context, warranting further investigation into these dynamics.

More work into the quality of juicy features, including audio and haptic feedback, could have been implemented given more time in the study. The current design relied heavily on visual feedback, but lacked extensive audio enhancements or varied haptic responses. These additional forms of juiciness could have been integrated using Unity's AudioSource and XR Controller APIs to test their impact on immersion and motivation, potentially amplifying the overall player experience. The time constraint limited this exploration.

Future research could focus on the specific effects of audio and haptic feedback on players compared to visual juiciness, addressing a gap in the current study. Investigating whether spatial audio or haptic responses enhance immersion and motivation could clarify juiciness's overall impact more broadly. This would build on the study's findings, offering deeper insights into optimising VR game design and tailoring feedback to specific player needs.

The study included 10 participants providing a solid foundation, however the small sample size and reliance on visual juiciness limited the study's scope. Future work could expand on the participant pool alongside a broader range of juiciness types to enhance the overall efficacy of the findings in this study.

Bibliography

- Abeele, V. V., Spiel, K., Nacke, L., Johnson, D. & Gerling, K. (2020), 'Development and validation of the player experience inventory: A scale to measure player experiences at the level of functional and psychosocial consequences', *International Journal of Human-Computer Studies* **135**, 102370.
- Ballou, N., Breitsohl, H., Kao, D., Gerling, K. & Deterding, S. (2021), Not very effective: Validity issues of the effectance in games scale, *in* 'Extended Abstracts of the 2021 Annual Symposium on Computer-Human Interaction in Play', pp. 55–60.
- Bodzin, A., Junior, R. A., Hammond, T. & Anastasio, D. (2021), 'Investigating engagement and flow with a placed-based immersive virtual reality game', *Journal of science education and technology* **30**, 347–360.
- Browning, M. H., Mimnaugh, K. J., Van Riper, C. J., Laurent, H. K. & LaValle, S. M. (2020), 'Can simulated nature support mental health? comparing short, single-doses of 360-degree nature videos in virtual reality with the outdoors', *Frontiers in psychology* **10**, 2667.
- Carter, J. (2024), Cmp3753 project assessment 1 16598945. Developed in partial fulfilment of the requirements for the Degree of BSc (Hons) Games Computing.
- Carter, J. (2025), Cmp3753 project assessment 2 16598945. Developed in partial fulfilment of the requirements for the Degree of BSc (Hons) Games Computing.
- Dozio, N., Marcolin, F., Scurati, G. W., Ulrich, L., Nonis, F., Vezzetti, E., Marsocci, G., La Rosa, A. & Ferrise, F. (2022), 'A design methodology for affective virtual reality', *International Journal of Human-Computer Studies* **162**, 102791.
- Durmanova, K. (2022), The effects of juicy game design on exergames, Master's thesis, University of Waterloo.
- Epic-Games (2025), 'fab.com'.
URL: <https://www.fab.com/>
- Fabre, É., Seaborn, K., Verhulst, A. A., Itoh, Y. & Rekimoto, J. (2024), Juicy text: Onomatopoeia and semantic text effects for juicy player experiences, *in* 'Proceedings of the 26th International Conference on Multimodal Interaction', pp. 144–153.
- Frommel, J., Klarkowski, M. & Mandryk, R. L. (2021), The struggle is spiel: On failure and success in games, *in* 'Proceedings of the 16th International Conference on the Foundations of Digital Games', pp. 1–12.
- Frost, S., Kannis-Dymand, L., Schaffer, V., Millear, P., Allen, A., Stallman, H., Mason, J., Wood, A. & Atkinson-Nolte, J. (2022), 'Virtual immersion in nature and psychological well-being: A systematic literature review', *Journal of Environmental Psychology* **80**, 101765.
- González-Cutre, D., Brugarolas-Navarro, M., Beltrán-Carrillo, V. J. & Jiménez-Loaisa, A. (2025), 'The frustration of novelty and basic psychological needs as predictors of maladaptive outcomes in physical education', *Physical Education and Sport Pedagogy* **30**(1), 64–77.

- Hicks, K., Dickinson, P., Holopainen, J. & Gerling, K. (2018), Good game feel: an empirically grounded framework for juicy design, *in* ‘Proceedings of DiGRA 2018 Conference: The Game is the Message’.
- Hicks, K., Gerling, K., Dickinson, P. & Vanden Abeele, V. (2019), Juicy game design: Understanding the impact of visual embellishments on player experience, *in* ‘Proceedings of the annual symposium on computer-human interaction in play’, pp. 185–197.
- Huang, W., Roscoe, R. D., Johnson-Glenberg, M. C. & Craig, S. D. (2021), ‘Motivation, engagement, and performance across multiple virtual reality sessions and levels of immersion’, *Journal of Computer Assisted Learning* **37**(3), 745–758.
- Kao, D. (2020), ‘The effects of juiciness in an action rpg’, *Entertainment Computing* **34**, 100359.
- Kao, D., Ballou, N., Gerling, K., Breitsohl, H. & Deterding, S. (2024), How does juicy game feedback motivate? testing curiosity, competence, and effectance, *in* ‘Proceedings of the CHI Conference on Human Factors in Computing Systems’, pp. 1–16.
- Li, L., Hew, K. F. & Du, J. (2024), ‘Gamification enhances student intrinsic motivation, perceptions of autonomy and relatedness, but minimal impact on competency: a meta-analysis and systematic review’, *Educational technology research and development* **72**(2), 765–796.
- Luz, A., Marcher, F., Nacke, L. E. & Vogel, D. (2024), Encouraging disengagement: Using eye tracking to examine attention with different levels of juicy design, *in* ‘Proceedings of the 2024 International Conference on Advanced Visual Interfaces’, pp. 1–5.
- Makantasis, K., Liapis, A. & Yannakakis, G. N. (2021), ‘The pixels and sounds of emotion: General-purpose representations of arousal in games’, *IEEE Transactions on Affective Computing* **14**(1), 680–693.
- Maroungkas, A., Troussas, C., Krouska, A. & Sgouropoulou, C. (2023), ‘Virtual reality in education: a review of learning theories, approaches and methodologies for the last decade’, *Electronics* **12**(13), 2832.
- Petersen, G. B., Petkakis, G. & Makransky, G. (2022), ‘A study of how immersion and interactivity drive vr learning’, *Computers & Education* **179**, 104429.
- Pichlmair, M. & Johansen, M. (2021), ‘Designing game feel: A survey’, *IEEE Transactions on Games* **14**(2), 138–152.
- Prithul, A., Adhanom, I. B. & Folmer, E. (2021), ‘Teleportation in virtual reality; a mini-review’, *Frontiers in Virtual Reality* **2**, 730792.
- Ribelles, J., Lopez, A. & Traver, V. J. (2024), ‘Modulating the gameplay challenge through simple visual computing elements: A cube puzzle case study’.
- Sheeran, P., Wright, C. E., Avishai, A., Villegas, M. E., Rothman, A. J. & Klein, W. M. (2021), ‘Does increasing autonomous motivation or perceived competence lead to health behavior change? a meta-analysis.’, *Health Psychology* **40**(10), 706.
- Shukla, D. (2020), ‘Ar and vr market size likely to grow exponentially’, *Market Survey, February* pp. 84–87.

- Singhal, T. & Schneider, O. (2021), Juicy haptic design: Vibrotactile embellishments can improve player experience in games, *in* ‘Proceedings of the 2021 chi conference on human factors in computing systems’, pp. 1–11.
- Smets, J. H. & Van der Spek, E. D. (2021), That sound’s juicy! exploring juicy audio effects in video games, *in* ‘Entertainment Computing–ICEC 2021: 20th IFIP TC 14 International Conference, ICEC 2021, Coimbra, Portugal, November 2–5, 2021, Proceedings 20’, Springer, pp. 319–335.
- Somarathna, R., Bednarz, T. & Mohammadi, G. (2022), ‘Virtual reality for emotion elicitation—a review’, *IEEE Transactions on Affective Computing*.
- Swink, S. (2009), *Game feel: A game designer’s Guide to Virtual Sensation*, CRC Press Taylor Francis Group.
- Tao, G., Garrett, B., Taverner, T., Cordingley, E. & Sun, C. (2021), ‘Immersive virtual reality health games: a narrative review of game design’, *Journal of NeuroEngineering and Rehabilitation* **18**, 1–21.
- Vanden Abeele, V., Spiel, K., Nacke, L., Johnson, D. & Gerling, K. (2020), ‘Development and validation of the player experience inventory: A scale to measure player experiences at the level of functional and psychosocial consequences’, *International Journal of Human-Computer Studies* **135**, 102370.
- Vo, T. T. D., Tuliao, K. V. & Chen, C.-W. (2022), ‘Work motivation: The roles of individual needs and social conditions’, *Behavioral Sciences* **12**(2), 49.
- Yang, S. & Zhang, W. (2022), ‘Presence and flow in the context of virtual reality storytelling: What influences enjoyment in virtual environments?’, *Cyberpsychology, Behavior, and Social Networking* **25**(2), 101–109.